Beyond the Dashboard: Business Intelligence as Sociomaterial Practice

A Qualitative Study on Role-Making and Sensemaking in Data-Driven Organizations

Master's Thesis 30 credits
Programme: Master's Programme in Business and Management
Specialisation: Organisation

Department of Business Studies
Uppsala University
Spring Semester of 2026

Date of Submission: 2026-05-27

Qiuchi Wang
Ziling Huang

Supervisor: Anna Tyllström

Abstract

In the current era of AI integration, data-driven organizations increasingly deploy Business Intelligence (BI) tools to optimize decision-making. However, while large-scale BI adoption is driven by expectations of enhanced efficiency, it frequently creates collaborative friction and disrupts established organizational dynamics. Rather than universally improving performance, the integration of these tools triggers a critical shift in organizational dynamics and reconfigures employee voice and agency. Existing literature largely covers the technical and performance-related outcomes of BI, less attention has been paid to these micro-level socio-technical tensions and role reconfigurations. Therefore, we aim to investigate how the implementation of BI tools actively reshapes internal organizational dynamics and redefines the interconnection between the tools and the people. To explore these dynamics, this study adopts a qualitative research design, drawing on 15 semi-structured interviews 15 semi-structured interviews with data practitioners and business managers within digitally mature sectors. The findings reveal that BI transforms dashboards into contested spaces where data logic clashes with experiential intuition. Furthermore, BI drives a definitive role reconfiguration, elevating data practitioners to decision gatekeepers while subjecting traditional managers to algorithmic control. Crucially, under rigid performance pressures, these tensions breed "data formalism," where actors manipulate metrics for risk deflection, decoupling data from genuine insights and highlighting the inherently political nature of corporate data work.

Key Words: Business Intelligence (BI); Organizational Sense-making; Decision-making; Role Reconfiguration; Sociomateriality; Data Formalism; Qualitative Study

Acknowledgments

First and foremost, we would like to express our heartfelt gratitude to our supervisor Anna Tyllström. Thank you for your selfless guidance and dedicated mentorship. In the midst of this long and unpredictable academic journey, your insights and wisdom have always been a beacon of light throughout our academic exploration.

Secondly, our sincere thanks go to our fellow classmates for their encouragement and companionship. The sparks of thought generated in every seminar were truly inspiring. Without everyone's open discussions, oversight, and support, this thesis could never have been continuously refined and brought to completion.

At the same time, we are deeply grateful to Uppsala University for providing an academic environment that is both rigorous and professionally stimulating, yet warm and inclusive. We also owe a special thank you to the beautiful and serene scenery outside the windows of Ekonomikum, during moments of creative blocks and exhaustion, looking out at the scenery was a source of inner peace and renewed strength.

Lastly, as this master's thesis comes to an end, we find ourselves approaching what may be the final graduation ceremony of our lives. Our gratitude to so many people and experiences goes far beyond words. What remains in our hearts is not merely academic theories, but the warm support and wisdom of our fellow classmates, alongside the invaluable mentorship of our teachers. Years from now, we hope to still savor the memory of this very moment, and remember the youthful, courageous spirits we once held. As the saying goes: "When we are young, all we have is a light heart and a body full of strength—but with just those two things, we can create the entire world."

Qiuchi Wang, Uppsala, 2026-05-27 Ziling Huang, Uppsala, 2026-05-27

Table of Contents

1. Introduction 1

2. Literature review and Theoretical Framework 3

2.1 Business Intelligence and Decision Support 3

2.2 BI as Sociomaterial Practice 4

2.3 BI and Organizational Sensemaking 5

2.4 BI Practices and Role Reconfiguration 6

2.5 Limitations of Existing Research 8

2.6 A Multi-Level Integrative Framework 8

3. Methodology 9

3.1 Research Design 9

3.2 Sampling Strategy and Informant Profile 10

3.2.1 Sampling Strategy 10

3.2.2 Informant Profile 10

3.3 Data Collection 11

3.3.1 Semi-structured Interviews 11

3.3.2 Ethical Considerations 12

3.4 Data Analysis 13

4. Empirical Findings 14

4.1 Thematic Map 14

4.2 Theme 1: Perceptions of the Value of BI Tools and the Boundaries of Their Limitations 15

4.3 Theme 2: The Role and Logic of Data in the Decision-Making Chain 17

4.4 Theme 3: Power Relations and Discursive Authority of Data Roles 19

4.5 Theme 4: Data Quality, Infrastructure, and Issues of Trust 21

4.6 Theme 5: The Evolution of Job Functions and Role Transformation 23

5. Discussion 25

5.1 BI as Sociomaterial Practice 25

5.2 BI and Organizational Sensemaking 26

5.3 BI Practices and Role Reconfiguration 27
5.4 Theoretical Contributions 28
5.6 Limitations and Future Research 30
6. Conclusion 31
References 33
Appendix 36
1. Semi-Structured Interview Outline 36
Phase 1: Icebreaker & Professional Background 36
Phase 2: Daily Collaboration & Role Division 36
Phase 3: Conflicts, Tensions & Decision Making 36
Phase 4: Deep Feelings & Identity 36
2. Informed Consent Form 37
1). Research Purpose and Description 37
2). Procedures 37
3). Confidentiality and Anonymity 37
4). Voluntary Participation and Right to Withdraw 38
5). Contact Information 38
6). Participant's Statement 38
3. Table 38
a. Code Frame 38
b. Code Book 44
c. Informant Profile 52

1. Introduction

In the increasingly digitalized business environment, organizations are overwhelmed with large amounts of data generated from internal operations and external market interactions. To address this complexity, Business Intelligence (BI) has evolved from a support tool to a central strategic asset, accounting for the largest share of global IT investment (Ransbotham & Kiron, 2017). Traditionally, BI is conceptualized as a technological capability that is a set of applications used to collect, access, and analyze data to enhance decision quality and organizational performance (Sabherwal & Becerra-Fernandez, 2011; Chen & Lin, 2021). The prevailing managerial logic often assumes a straightforward process: better data infrastructure automatically leads to better decisions (Jordan & Ellen, 2009). However, this technology-biased view often treats the decision-making process as a mechanical calculation, overlooking the complex organizational dynamics through which data is actually interpreted and utilized.

Critically, a central element of these organizational dynamics is that decision-making is rarely a purely rational process of selecting alternatives. Instead, decision-making serves as a broader context that presupposes a fundamental process of sensemaking, in which actors must first construct meaning from ambiguous cues before any action can be taken (Boland, 1984; Namvar et al., 2016). Weick (1995) further conceptualizes this as an interpretive process embedded in a social context, where individuals draw on prior experiences to produce plausible accounts of reality. In this view, BI systems do not merely provide objective answers; rather, they serve as inputs for an ongoing debate over meaning. Maitlis and Lawrence (2007) and Kaplan (2008) demonstrate that this construction involves sensegiving and framing contests, highlighting that interpretation is deeply shaped by organizational roles and authority structures.

Furthermore, Technological systems also shape these participation structures. Technologies are not only supportive tools but also embedded within organizational arrangements and can change how people interact and coordinate in important ways (Orlikowski, 1992; Leonardi, 2011). Since different groups may understand the same system differently (Orlikowski & Gash, 1994), the introduction of BI practices does more than automate tasks. It also creates new forms of visibility and control, which can challenge established hierarchies. Therefore, in this study, organizational dynamics are conceptualized mainly through two closely related processes: sensemaking and role configuration. Decision-making is treated as the broader context in which these dynamics develop and become visible.

Although the strategic value of BI has been widely discussed, there is still a clear gap in understanding how these technologies interact with internal organizational dynamics. Mekimah et al. (2024) provide a bibliometric review of BI research over the past decade. Their review shows that the connection between BI, decision-making processes, and human resource management remains underexplored. Based on this broader gap, this study focuses on a more specific issue: how the two core dimensions of organizational dynamics, sensemaking and role configuration, interact and develop within an organization.

First, rational decision-making models often give limited attention to the early stage of sensemaking (Fink et al., 2017; Božič & Dimovski, 2019). Before an organization can act on data, employees need to build a shared understanding of the ambiguous signals from BI dashboards. Some studies, such as Namvar et al. (2016), suggest that BI can support this process. However, we still know little about the micro-level interactions involved. This is especially important when organizational employees need to negotiate and reach consensus when the data is unclear or conflicts with their experience.

Additionally, the introduction of BI often changes existing structures and leads to important role changes. However, this process is rarely studied together with how people interpret data. Modern BI systems create greater visibility, which can challenge traditional boundaries and change who has the authority to interpret information. Shollo (2013) shows that BI redefines the relationship between IT engineers and business managers. However, existing literature does not fully explain how this dynamic process of role-making shapes and is shaped by organizational sensemaking.

Finally, sensemaking and role changes are usually studied separately. As a result, we still lack a complete understanding of how these two processes develop together and shape the broader decision-making process. When BI metrics conflict with a manager's traditional experience, tensions may emerge in both sensemaking and role authority. These tensions can make decision-making more defensive and contested. In some cases, this may lead to "data formalism," where data is selectively used for symbolic or political purposes to maintain appearances rather than to generate genuine insights (Shollo, 2013).

To address these gaps, this study draws on three theoretical frameworks. First, sociomateriality helps us understand BI not as a neutral tool, but as part of organizational practice (Orlikowski & Scott, 2008). Second, Weick's sensemaking framework explains how

actors construct meaning from data-driven cues (Weick, 1995). Third, role dynamics theory and structuration perspectives provide concepts for examining how BI-related changes shift interpretive authority (Bosworth & Kreps, 1986; Ranson et al., 1980).

Based on these perspectives, this study moves beyond a purely rational view of decision-making. It adopts a social interaction approach to examine the ways in which BI shapes organizational dynamics. Therefore, this study proposes the following main research question:

RQ: How do BI (business intelligence) practices reconfigure organizational dynamics in data-driven organizations?

Specifically, we conceptualize organizational dynamics through sensemaking and role reconfiguration. This study aims to develop a multi-level integrative framework showing how BI reshapes these dynamics within the broader context of organizational decision-making. This will help explain why the same BI tools can produce such different outcomes in different organizations, and how companies can turn raw data into shared, actionable insights.

The remainder of this thesis is structured as follows. Chapter 2 reviews the relevant literature on Business Intelligence, organizational sensemaking, and the role of participation in decision-making processes. Chapter 3 outlines the qualitative multi-case study methodology and describes the data collection conducted within digitally mature sectors in China. Chapter 4 presents the empirical findings. Chapter 5 discusses the empirical findings in relation to the theoretical framework, while outlining theoretical and practical implications alongside directions for future research. Chapter 6 concludes the thesis by returning to the central research question, synthesizing the empirical evidence to demonstrate how BI acts as a disruptive social force within data-driven organizations.

2. Literature review and Theoretical Framework

2.1 Business Intelligence and Decision Support

Business Intelligence (BI) has been conceptualized in different ways depending on the emphasis of the scholar. From a technological perspective, Abusweilem and Abualoush (2019) describe it as a set of applications and technologies used to collect, access, and analyze data in order to assist in making effective business decisions. Olszak (2014) takes a broader view, arguing that BI is an umbrella term that combines architectures, tools, databases, analytical

tools, applications, and methodologies to provide actionable information. What these definitions share is an orientation toward decision support, a shift that Davenport (2010) captures, observing that the dominant logic of enterprise information management has moved away from automating transactions and toward supporting decisions. This evolution marks a significant change in how organizations are expected to relate to their data.

Despite this strategic prominence, the organizational adoption of BI dashboards has rarely yielded uniform outcomes, with decision-making efficiency and cognitive consequences varying considerably across settings even when the underlying technological tools are identical (e.g., Davenport, 2010; Suša Vugec et al., 2020). Davenport (2010) attributes this variance in part to the loosely coupled relationship between information and decision-making, noting that the mere provision of data rarely guarantees consensus and can, conversely, trigger cognitive overload rather than clarity. Similarly, Suša Vugec et al. (2020) demonstrate that BI's value creation is largely indirect, relying heavily on its integration with broader organizational processes. However, as Orlikowski (2000) argues, technological tools do not produce outcomes in a social vacuum; rather, their consequences are fundamentally shaped by how different organizational groups interact with them in practice. Furthermore, because different roles often hold disparate interpretive frameworks regarding the same system (Orlikowski & Gash, 1994), the deployment of identical BI tools inevitably triggers varying degrees of social friction, communication barriers, and structural adjustments.

This critical gap between standardized technological tools and highly heterogeneous organizational results remains undertheorized in current literature. As contemporary information systems research suggests, explaining this variance requires moving beyond a purely techno-centric perspective to contextualize technology within the micro-level dynamics of an organization (e.g., Orlikowski, 2000). Consequently, examining how BI practices interlock with the underlying social structures, formalized roles, and shifting meaning-making processes through which organizations function becomes essential to understanding the reality of technology-driven decision support.

2.2 BI as Sociomaterial Practice

To gain a deeper understanding of how Business Intelligence (BI) integrates into organizations, we introduce the theory of sociomateriality. Orlikowski and Scott (2008) challenge the traditional assumption of studying technology and organization separately,

proposing a constitutively entangled relationship between the two. This perspective argues that technology, work, and organization are inherently inseparable, meaning that technological practices are always dynamic sociomaterial configurations. Applying this lens to data-driven environments implies that BI practices can no longer be viewed merely as independent technological tools. Rather, when an organization introduces BI, this material force inevitably interacts with the existing social structure, with its operational consequences varying based on localized practices and organizational goals. Therefore, transforming BI into an effective organizational capability depends heavily on specific socio-technical alignment. Prior research identifies key decisive components within this framework, emphasizing that technical infrastructure, management commitment, and a data culture of fact-based decision making must be reasonably well aligned for BI to function as intended (Olszak, 2016; 2022).

This socio-technical alignment points to a deeper, symbiotic relationship between BI practices and organizational decision-making processes. As Orlikowski (1992) posits through the concept of the duality of technology, technological artifacts do not merely serve tasks; instead, they actively structure ongoing human interactions, reinforcing or reshaping established institutional rules and communicative routines. Viewed through this lens, BI practices do more than passively deliver analytical inputs. The implementation of dashboards reconfigures the structural distribution of information and alters the patterns of organizational visibility (e.g., Leonardi, 2011). By intervening in who possesses data and where information generates legitimacy, BI practices actively transform the operational foundations of an organization. In this sense, the technology serves as an active sociomaterial force that reshapes organizational dynamics, effectively setting the structural boundaries within which the micro-level processes of human sensemaking and role reallocation unfold.

2.3 BI and Organizational Sensemaking

The sensemaking framework developed by Weick (1995) offers a valuable lens for understanding how organizational members construct meaning from the ambiguous signals produced in data-driven environments. Rather than treating decision-making as a rational computation among known alternatives, Weick conceptualizes it as an ongoing interpretive accomplishment, where actors rely on prior experiences, social interactions, and extracted cues to produce accounts of reality that are plausible enough to support action. Crucially, because this interpretive process is inherently social — emerging through continuous negotiation across organizational roles — and driven by plausibility rather than absolute

accuracy, the implementation of BI systems does not simply deliver mechanical answers. Instead, it injects new material cues into an already dynamic and ongoing process of meaning construction.

In contemporary data-dense environments, managing the cognitive load generated by vast volumes of ambiguous information has become a central organizational challenge. Namvar et al. (2016) argue that BI systems can play a crucial scaffolding role in this process, supporting the extraction of meaningful cues through automated reporting and real-time monitoring. Bačić and Fadlalla (2016) extend this argument by examining how business information visualization reshapes cognitive engagement with data, suggesting that visual representations do more than merely present figures; they actively shape which patterns decision-makers notice and how they interpret what they see. Together, these perspectives demonstrate that BI practices do not merely accelerate information flow. Rather, they directly intervene in the interpretive filters through which organizational reality is collectively constructed, generating a persistent cognitive tension between data-driven cues and experiential judgment that this study seeks to examine empirically.

2.4 BI Practices and Role Reconfiguration

Within traditional organizational configurations, roles are predominantly conceptualized as a manifestation of stable social order. Bosworth and Kreps (1986) describe this as a state of role-playing, in which organizational actors derive their behaviors and expectations from established rules and a consistent status-role nexus. This portrays the division of labor as defined and fixed, relying heavily on historical routines and structural continuity. Meyer and Rowan (1977) offer a foundational explanation for how such stability persists, arguing that formal structures often function as institutional myths that are decoupled from actual technical activities, thereby shielding daily routines from external scrutiny.

The proliferation of BI technologies, however, disrupts this established equilibrium. As Orlikowski (1992) and Leonardi (2011) argue, technology functions as an active agent that reshapes interaction patterns. Specifically, the data capabilities enabled by BI introduce an inconsistency into the traditional status-role nexus by bridging the gap between formal institutional structures and actual workplace practices. This structural disruption propels organizational members away from static scripts and toward what Bosworth and Kreps (1986) define as role-making—a highly improvisational process in which actors dynamically enact new roles contingent upon an evolving data context.

This dynamic role-making process is fundamentally driven by two interrelated socio-technical mechanisms: enhanced behavioral visibility and the shift toward algorithmic control. On one hand, BI systems drastically increase behavioral visibility by capturing granular digital traces, creating what Leonardi and Treem (2020) term "network translucence". This heightened transparency reveals that operational expertise often resides not within formal hierarchical leadership positions, but with technical actors who manipulate and interpret data (Schafheitle et al., 2020). Consequently, technical specialists handling data gain new forms of localized authority, allowing them to expand their role boundaries beyond conventional back-office support functions in ways that challenge established organizational hierarchies.

On the other hand, BI practices alter the fundamental mechanisms of organizational control by shifting management from traditional human supervision toward automated algorithmic governance. In this setting, BI systems evolve from merely descriptive tools into prescriptive engines, transforming quantitative indicators into rigid behavioral standards that constrain managerial discretion (Schafheitle et al., 2020). Traditional managers are thus compelled to reconfigure their roles from sovereign commanders into interpreters of algorithmic logic. However, this transition is rarely straightforward. Plesner and Justesen (2025) introduce the concept of "techno-organizational rescripting" to emphasize that algorithms are context-dependent entities whose organizational effects remain contingent upon the specific scripts into which they are embedded. Their findings demonstrate that employee roles are simultaneously multiplied and restricted, suggesting that the reconfiguration of managerial authority is better understood as a contested, context-sensitive negotiation rather than a seamless or predetermined transformation.

These converging shifts in visibility and control ultimately culminate in what Ranson et al. (1980) describe as the reconfiguration of interpretive authority. Drawing on a structuration perspective, Ranson et al. (1980) argue that organizational structures are constituted by power dependencies grounded in specific provinces of meaning. In BI-driven environments, data analytics emerges as the dominant province of meaning for constructing institutional reality. As Leonardi and Treem (2020) indicate, data visibility affords new forms of influence to those who possess information, meaning that organizational authority no longer adheres solely to formal hierarchical status. Instead, it flows toward actors who can effectively bridge technical data logic and business context, fundamentally altering not only who is entitled to speak in decision-making but also the very sources of legitimacy through which organizational reality is collectively constructed.

2.5 Limitations of Existing Research

Existing literature provides a foundational understanding of the evolution and strategic value of Business Intelligence (BI). As Davenport (2010) notes, the management paradigm has undergone a significant shift from transactional automation to decision support, with subsequent strategic research demonstrating BI adoption as an important driver of organizational performance. These traditional studies have successfully contributed to elevating BI from a mere technical tool to a recognized organizational capability.

However, much of this research remains grounded in a functionalist perspective, which tends to assume that organizational actors will rationally follow data insights once the technology is implemented. This deterministic view frequently overlooks a central organizational challenge identified by Myth and Ceremony theory (Meyer & Rowan, 1977): formal structures, such as BI platforms, often exist as institutional myths to secure external legitimacy, while in practice remaining deeply decoupled from actual daily decision-making activities. While existing BI literature extensively discusses the functional merits of technology, it does not sufficiently explain how this strategic decoupling affects micro-level execution, or how organizations navigate internal structural tensions when data-driven transparency is mandated.

Furthermore, although recent scholars has begun to address the digital impacts of behavioral visibility (Leonardi & Treem, 2020) and algorithmic control (Schafheitle et al., 2020), these insights remain highly fragmented. Current studies have not yet brought these technological features into dialogue with one another in a way that explains how they collectively reshape the social reality of work。 Existing BI research still largely treats technology as an exogenous variable separate from human action. Crucially, current literature lacks the integrative application of structuration perspectives (Ranson et al., 1980) and role dynamic frameworks (Bosworth & Kreps, 1986). A notable gap remains in tracing how sociomaterial entanglement, through ongoing micro-level role interactions, ultimately translates into a structural reconfiguration of interpretive authority within data-driven organizations.

2.6 A Multi-Level Integrative Framework

Drawing deductively from the theoretical perspectives reviewed above, we propose a multi-level integrative framework that conceptualizes BI not merely as an analytical tool, but as a disruptive sociomaterial force that triggers a comprehensive structuration process. Each stage of this framework follows logically from the theoretical building blocks established in the preceding sections.

The framework establishes its baseline state upon the institutional perspective of Meyer and Rowan (1977), where traditional organizations rely on decoupling mechanisms to maintain a stable social order grounded in static role-playing. In this initial phase, formal structures, including BI systems when first deployed, frequently function as institutional myths disconnected from actual daily execution. This decoupled stability is subsequently disrupted by the technological trigger mechanisms inherent in BI practices. By enhancing behavioral visibility (Leonardi & Treem, 2020) and enforcing systemic algorithmic control (Schafheitle et al., 2020), BI compels organizational actors to confront the acute cognitive tensions between data-driven metrics and traditional experiential intuition. This structural disruption, in turn, drives an active process of micro-level role reconfiguration. Organizational members are forced beyond static formalized scripts and propelled into the dynamic, improvisational "role-making" described by Bosworth and Kreps (1986), actively navigating the underlying friction between established professional habits and emerging data logics.

The outcome of this process, drawing on Ranson et al. (1980), is a redistribution of interpretive authority within the organization. Ultimately, this framework discusses how BI catalyze a shift in organizational dynamics by reshaping both who is entitled to participate in decision-making and the sources of legitimacy through which organizational reality is collectively constructed.

3. Methodology

3.1 Research Design

To address the research question — how do BI practices reconfigure organizational dynamics across different industries and organizational roles — we adopt a qualitative research design. Qualitative approaches are particularly suited to this inquiry because organizational decision-making is rarely a purely rational computation but a complex social process involving interpretation, negotiation, and contestation. As Yin (2018) argues, qualitative case study designs are optimal for answering "how" questions that examine contemporary phenomena in real-world contexts. Furthermore, this approach allows us to view BI systems not simply as neutral tools but as active sociomaterial agents (Orlikowski & Scott, 2008), making it possible to observe empirically how their material properties interact with human practices to trigger dynamic role-making (Bosworth & Kreps, 1986).

Within this qualitative orientation, we draw on a multi-case logic to strengthen the analytical reach of the findings. Instead of seeking statistical generalizability, a multi-case design enables the identification of patterns and variations across different organizational and industry contexts, which is central to addressing the research question (Eisenhardt, 1989; Eisenhardt & Graebner, 2007). A total of 15 semi-structured interviews were conducted with informants occupying a range of roles across multiple industries in China, including internet platforms, financial technology, traditional finance, real estate, smart home appliances, and retail. By sampling across both technical and business-oriented roles, we aimed to capture the social friction and interpretive negotiation that occurs at the boundary between data logic and experiential judgment. The detailed procedures for sampling, data collection, and data analysis are presented in the sections that follow.

3.2 Sampling Strategy and Informant Profile

3.2.1 Sampling Strategy

We adopt a purposive sampling approach in the empirical study, selecting informants who are likely to provide rich and varied perspectives on the research question. Participants were recruited through a combination of the researchers' professional networks and a public call for participation posted on Xiaohongshu, a Chinese social media platform widely used by community. The primary criterion for inclusion was that informants had substantial experience working with BI systems in a professional capacity, either as data or technical specialists or as business managers and decision-makers. A secondary criterion was that the sample should reflect variation across industries and organizational roles, in order to support the study's aim of examining how BI practices reconfigure organizational dynamics across different contexts.

To achieve this diversity, the 15 informants were selected from the aforementioned sectors, ensuring a balanced cross-industry representation. In terms of functional role, eight informants were positioned on the business and decision-making side, including product managers, operations leads, strategy managers, and commercial analysts, while seven occupied technical or data-oriented roles, including BI analysts, data engineers, and risk management specialists. The sample size of fifteen is consistent with qualitative norms, where theoretical saturation rather than statistical representativeness governs adequacy (Eisenhardt, 1989).

3.2.2 Informant Profile

Table 1 presents the profile of all fifteen informants. To protect participant confidentiality, all informants are referred to by codes from S01 to S15 throughout this study. The sample includes informants from organizations ranging in size from approximately 300 to over 10,000 employees, and BI experience levels ranging from moderate project-based use to deep architectural engagement.

Table 1. List of Informant Profile

ID	Job Title	Decision Role	Industry	Company Size	BI Experience
S01	Digital Product Manager	Platform Builder / Data Advisor	Internet	10,000+	Heavy / Architecture-Level User / Heavy (Developer & Analytics Expert)
S02	Internet Business Operations	Demand Initiator & Decision Advisor	Internet / E-commerce	1,000+	Heavy (Business Loop-Driven) / Hands-on
S03	UX Product Manager	Demand Initiator & Stakeholder Manager	Internet	1,000+	Moderate / Project-Based
S04	BI Data Analyst	Defensive Decision Supporter / Metric Controller	Traditional Finance / Non-Auto Insurance Claims	10,000+	Moderate / Reluctant Adopter
S05	Senior Business Analyst (BA)	Organizational Bridge & Third-Party Observer	Internet / Multi-Business Platform	10,000+	Deep / Architecture-Level User
S06	Business Analyst	Middle-Platform Decision Supporter	Internet / F&B Supply Chain	10,000+	Moderate-Heavy (Analysis-Oriented)
S07	Middle-Platform Operations Analyst	Goal Setter & Solution Provider	Internet Live-Stream E-commerce	10,000+	Heavy / High-Frequency User
S08	Head of Risk Control	Risk Gatekeeper / Decision Participant	FinTech	300+	Heavy / Self-Service User
S09	BI Product Manager	Operations Decision Maker / End-to-End Owner	Whole-Home Intelligence	10,000+	High-Frequency / Hybrid User
S10	Business Analyst	Analytics Middle-Platform & Joint Decision Maker	Internet Mobility Platform	10,000+	Heavy / Expert-Level
S11	Product & User Operations	Business Decision Maker & Dashboard Builder	Internet Mobility Platform	10,000+	Heavy (Hands-on)
S12	Strategy Lead	Business Decision Maker & Demand Definer	Internet Mobility Platform	10,000+	Heavy (Strategy Monitoring Perspective)
S13	Data Operations Manager	Decision Supporter / Data Interpreter	Real Estate	10,000+	Heavy (Senior Analytics Perspective)
S14	Data Analyst	Decision Supporter & Tool Builder	Internet	10,000+	Heavy / Expert-Level
S15	Big Data Development Engineer	Technical Builder / Decision Supporter	Retail	300+	Heavy / Developer & Monitor

3.3 Data Collection

3.3.1 Semi-structured Interviews

Semi-structured interviews serve as the primary method for data collection in this study. This format provides the flexibility to explore complex social dynamics while ensuring that core theoretical themes are consistently addressed across informants (Bryman, 2016). A total of fifteen interviews were conducted, each lasting approximately 30 to 45 minutes. All interviews were conducted online via video call, a format that proved practical given the geographical spread of informants across different cities in China. With participants' consent,

all interviews were audio-recorded and subsequently transcribed verbatim for analysis. Given that all informants are based in China, interviews were conducted in Mandarin to ensure naturalness and depth of expression. To ensure conceptual equivalence and linguistic accuracy during the data preparation phase, advanced artificial intelligence (AI) translation tools were utilized to generate the initial English scripts, which were subsequently cross-verified and corrected by the researchers. The finalized transcripts were then used for analysis.

The interview protocol is structured around four thematic blocks that directly mirror the conceptual framework developed in Chapter 2, moving from professional background and daily work practices, through moments of tension and conflict, to broader reflections on role identity and organizational change. Each block opens with a substantive question followed by targeted follow-up probes designed to elicit concrete incidents and lived experiences rather than abstract generalizations. The protocol begins with a warm-up stage inviting participants to describe their current role and their team's positioning within the organization, establishing rapport before entering more substantive territory.

The core of the protocol focuses on how BI shapes collaboration and decision-making in practice. Participants were asked to reflect on specific moments where data outputs and experiential judgment pulled in different directions, and to describe how such tensions were navigated and whose interpretation ultimately carried weight. This line of questioning is central to the study's interest in the social friction that emerges at the boundary between data logic and business intuition. The protocol closes with reflective questions about role identity and organizational change, including how participants perceive their own authority within the decision-making chain, and what they imagine would happen to their work if the BI system were suddenly unavailable. Where relevant, questions were adapted to reflect the specific vantage point of each informant, with technical specialists asked more about how their analysis influenced decisions, and business managers asked more about how they navigated conflicts between system outputs and their own judgment.

3.3.2 Ethical Considerations

This study was conducted in accordance with standard ethical guidelines for qualitative research (Bryman, 2016). Prior to recruitment, all potential participants were informed of the research objectives, the voluntary nature of participation, and their right to withdraw at any point without consequence. Given the online recruitment context, informed consent was obtained through an implied consent procedure: participation in the interview was taken to

indicate agreement with the stated terms, which were communicated clearly in the recruitment post and reiterated at the start of each interview. No formal written consent form was required, as this approach is considered appropriate for low-risk research of this nature (Creswell & Creswell, 2018).

To protect participant confidentiality, all informants are referred to by anonymized codes throughout the study, and no identifying information is reported in the findings. Interview recordings and transcripts are stored securely and used exclusively for academic purposes. Participants were assured that their data would not be shared with third parties or used for commercial gain.

3.4 Data Analysis

To analyze the interview transcripts, we follow the six-phase thematic analysis process outlined by Braun and Clarke (2006): familiarization with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final report. This approach allows the analysis to remain grounded in empirical data while being informed by the theoretical framework developed in Chapter 2.

In the first phase, all interview transcripts were read repeatedly to develop thorough familiarity with the data. Given that interviews were conducted in Mandarin, an initial translation process was applied to ensure conceptual equivalence before coding commenced. During this data-cleaning stage, AI-assisted text processing was employed to refine grammatical structures, eliminate conversational redundancies (such as filler words), and polish the academic phrasing of the translated text without altering the core meaning or empirical substance of the informants' accounts. Coding then commenced on the translated transcripts, with initial attention given to passages relating to the four thematic areas of the interview protocol: professional background and daily work practices, collaboration and role division, tensions and decision-making moments, and reflections on role identity and organizational change.

Following transcription, the data were segmented into meaning units, with each unit corresponding to a coherent idea or experience expressed by a respondent, regardless of where it appeared within the interview. Each meaning unit was then assigned a descriptive label capturing its substantive content. This initial labeling process produced approximately 488 descriptive labels across the fifteen interviews. These labels were subsequently reviewed and grouped through iterative discussion between the two researchers, with similar labels

collapsed into broader codes. A total of 32 codes were generated in this way, which were then organized into the five themes presented in Chapter 4. The full coding framework, including representative data excerpts, descriptive labels, codes, and themes, is provided in Appendix 3.a, and detailed code definitions are provided in Appendix 3.b.

To strengthen the credibility of the analysis, we adopted two additional measures. First, peer debriefing was employed through regular discussions with the research supervisor to challenge emerging interpretations and guard against researcher bias. Second, given that the study involves two researchers, a collaborative coding process was adopted, in which both researchers discussed and compared their interpretations of the data at regular intervals throughout the analysis. Disagreements over the application of codes or the boundaries of themes were resolved through discussion until consensus was reached. This process helped ensure consistency in how codes and themes were defined and applied across the dataset.

4. Empirical Findings

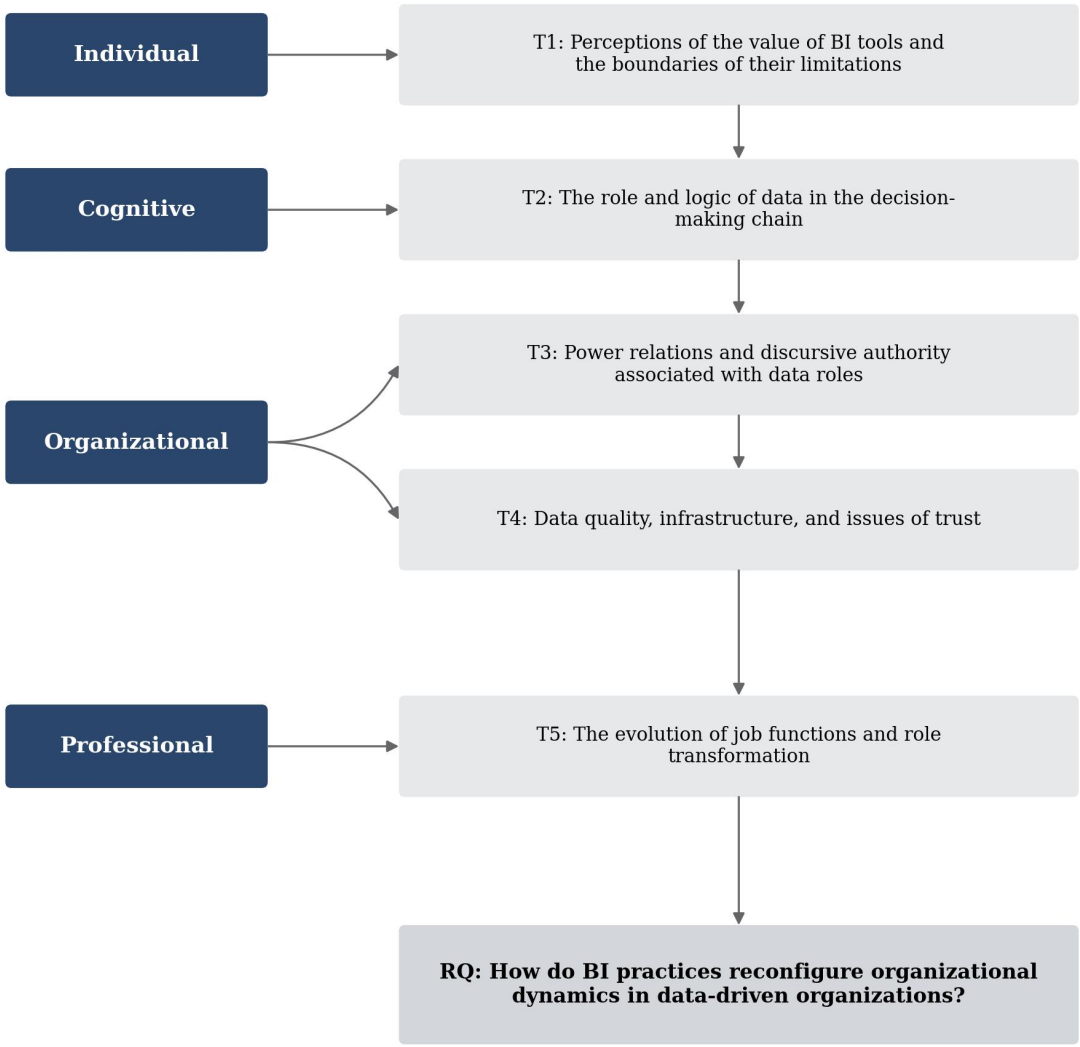
4.1 Thematic Map

This chapter presents the empirical findings of the study, organized around five themes that emerged from thematic analysis of fifteen semi-structured interviews conducted with data practitioners and business managers across different industries and organizational roles in China. The analysis generated 32 codes from an initial set of approximately 488 descriptive labels drawn from the fifteen interviews; the full coding framework is provided in Appendix 3.a. Together, these themes address the research question: How do BI practices reconfigure organizational dynamics in data-driven organizations?

The five themes are organized across four analytical layers that move from the individual experience of using BI tools to broader questions about organizational power and professional identity. The first two themes are concerned with how BI shapes the way individuals perceive and reason with data, including how it is relied upon, where it falls short, and how it interacts with experiential judgment in practice. The third and fourth themes shift to the organizational level, where questions of authority, collaboration, and data governance become more prominent. The fifth theme sits at the intersection of individual and organizational concerns, touching on how the changes described in earlier themes appear to be reshaping professional roles and career trajectories in ways that are still unfolding. While these layers are presented sequentially, they are not intended to suggest a strict causal progression. Instead, they reflect

different dimensions of a shared set of organizational processes that the data points toward. This structure is illustrated in Figure 4.1.

Figure 4.1. Research Thematic Map



Overall, the following five themes do not describe isolated phenomena but a connected set of organizational processes through which BI practices reshape how people work, how decisions are made, and how roles are defined and contested. Each theme is presented in turn in next sections.

4.2 Theme 1: Perceptions of the Value of BI Tools and the Boundaries of Their Limitations

This theme primarily explores organizational members' subjective assessments of the utility of Business Intelligence (BI) tools across various industries and roles, alongside the instrumental limitations encountered in complex business practices. It reflects how technology

has transformed from a peripheral, passive means of data extraction into an indispensable organizational infrastructure, while simultaneously, this deep technological embedding has prompted organizations to reflect on the experiential dead zones that data cannot reach.

Logically, the four core codes within this theme present a progressive relationship from "technological empowerment" to "organizational dependency," ultimately leading to "reflection on limitations". The starting point of BI's value lies in its explicit attributes of "reducing labor costs and improving analytical efficiency". This leap in efficiency makes high-frequency "routine monitoring of core metrics" possible and solidifies it as an organizational habit. As BI becomes deeply embedded in the business lifeline, organizations develop "the fear of operational standstill without BI". However, when dealing with complex business logic, users gradually perceive the system's "data blind spots and limitations of prediction," thereby establishing the boundaries of technology's application.

Firstly, respondents generally noted that the most direct value of BI lies in achieving data automation and freeing up human resources, which provides high-efficiency raw materials for subsequent decision-making. This efficiency improvement significantly reduces reliance on manual data pulling. One respondent pointed out: "The first and most direct benefit is efficiency. Without BI, someone would have to extract the same data every day, which consumes manpower and creates repetitive work" (S05). This technological empowerment allows organizations to maintain a high-frequency monitoring rhythm, with respondents describing their daily habit as: "Every morning, the first work habit is to monitor whether data fluctuations align with expectations through core indicator dashboards" (S05). This empirical evidence demonstrates that by solidifying trivial data operations into automated processes, BI not only achieves analytical efficiency but also reshapes the starting point of organizational members' daily work.

Secondly, because this efficient monitoring capability has been deeply integrated into the organization's lifeblood, a strong path dependency on BI has emerged, such that the system's absence would lead to a severe operational crisis. Respondents generally hold a fear of standstill regarding the scenario of "losing BI." Data presents itself as "indispensable," with a respondent confessing: "If BI is stopped, daily operation and supervision will directly face paralysis" (S09). This dependency even manifests as a loss of basic capabilities; if system support were lost, the team would be forced to "revert to manual spreadsheets" (S13) , resulting in a lack of perception, "as if being blinded". This set of arguments suggests that BI

tools have evolved from simple software applications into a sociomaterial configuration that supports the organization's daily operations; once removed, the original decision-making order would collapse.

Finally, although BI tools bring significant monitoring convenience, their instrumental limitations become exceptionally apparent when handling non-standardized, sudden, and high-granularity business activities. Users clearly recognize that data cannot fully reconstruct business reality and that obvious cognitive blind spots exist. A respondent warned: "Excessive reliance on data may create blind spots, because data cannot fully capture sudden market changes or the real experiences of users in highly segmented and personalized scenarios" (S02). Especially in complex offline interaction scenarios, respondents criticized BI systems for their inherent bias toward presenting aggregated averages, making it difficult to reflect the localized process value. As S09 mentioned:" BI systems naturally lean toward presenting aggregate averages and find it hard to capture partial structures or relatively complete process details." This experiential perception of limitations suggests that BI tools have real limitations; they still face persistent experiential limitations in dealing with the certainty of predicting the future and capturing micro-level user feelings.

4.3 Theme 2: The Role and Logic of Data in the Decision-Making Chain

This theme explores how data, no longer merely static reporting material, is deeply embedded in and reshapes the entire decision-making chain within an organization. It reflects a paradigm shift from traditional intuition-led models to a composite data logic, revealing the complex interaction between objective data and human experience in the decision-making process.

Under this theme, the various codes present a dynamic and progressive logical chain: first, the starting point of decision-making is front-loaded to utilizing BI systems for "discovering anomalies and pinpointing problems" and "attributing fluctuations and dissecting links"; based on these quantified facts, organizations evolve new models of "supporting decisions and logical endorsement" and "strategy iteration and A/B testing"; however, this rational data logic inevitably experiences friction with "experience intuition and subjective judgment", thereby triggering the negotiation of "data validation and overturning experience"; ultimately, organizations realize that if "data detached from real business contexts" occurs, its value to decision-making is significantly diminished, and only by combining data with "business insights and deepening logic" can true scientific decision-making be achieved.

At the front end of the decision-making chain, the monitoring and attribution capabilities of the system appear to reshape the trigger mechanisms for organizational action. Respondents indicated that the current standard workflow involves getting a global overview first: "first pull full-volume core metrics via BI to clarify the current situation" (S08). When metric anomalies are detected, the system assists the team in conducting deep horizontal and vertical analyses. For example, business personnel will drill down through multi-dimensional data to identify "anomalies in historical claims" (S04) or "abnormally high rental rates" (S15). This front-loaded process of "discovering anomalies and pinpointing problems" ensures that the starting point of decision-making is built upon objective facts, substantially shortening the response cycle from problem detection to taking action.

Based on precise attribution, data begins to play a central role of "supporting decisions and logical endorsement" in strategy formulation. In highly digitized environments, data provides a quantitative basis for business directions, making decision-making more agile. Respondents noted that by setting "objective data standards and fixed report dimensions," organizations have greatly "reduced the occurrence of arbitrary, head-slapping decisions" (S15). Furthermore, "strategy iteration and A/B testing" has become a standardized method for validating the effectiveness of decisions, making "the cycle from idea to validation" shorter and more efficient (S12). However, in certain organizations lacking a data-driven culture, data is sometimes alienated into a tool for finding evidence for pre-established decisions, leading to the inverted phenomenon where they "make the decision first, and then look for data tools" (S03).

As data deeply intervenes in decision-making, the emerging data logic often comes into tension with deeply ingrained "experience intuition and subjective judgment." When data outcomes are inconsistent with established experience, different roles exhibit their own defensive and validation logics: the business side tends to prioritize questioning "whether the data parameters and experimental design are flawed" (S12), while the data side focuses on checking the underlying logic. However, once subjected to rigorous "data validation and overturning experience," data demonstrates persuasive power (S02). One respondent admitted that solid data analysis challenged their subjective intuition that "high-value users need large coupons," prompting the team to establish a code of conduct based on "relying on data, not blindly trusting experience" (S02).

Although data can carry considerable weight in the validation phase, respondents still emphasized that entirely relying on data while ignoring "business insights" may carry decision-making risks. Especially when "data detached from real business contexts" occurs, a pure numbers game cannot answer the underlying commercial logic. Respondents explicitly pointed out that data often only reflects what has happened on the surface, while uncovering deep-seated reasons and providing risk safety nets must still "rely on business experience" (S08). If the analysis is divorced from specific application scenarios, it is as impractical as "castles in the air" lacking a foundation (S11). This suggests that BI practices do not eradicate human experience, but rather drive organizations toward a composite decision-making ecosystem where "data sets the direction, experience finds the reason" (S02).

4.4 Theme 3: Power Relations and Discursive Authority of Data Roles

This theme primarily explores the reconfiguration of power dynamics among different internal organizational roles under the influence of BI practices. It focuses not only on the ownership of decision-making authority but also on how data transparency empowers specific roles with discursive authority and how this expansion of power negotiates and compromises with traditional experiential authority.

Logically, the codes within this theme present an evolutionary process from "power empowerment" to "boundary friction" and finally to "collaborative symbiosis": First, the popularization of BI tools directly drives the "elevation of discursive power and identity advancement" among respondents. However, this expansion of power can encounter "experience barriers of business operations." In the collision between the two, organizations establish a "division of final decision-making power" to maintain systemic stability. Ultimately, this reconfiguration of power relations leads to a new partnership characterized by the "complementarity of business and data experts."

Firstly, the technological visibility afforded by BI appears to enhance the discursive authority of data-related roles in organizational games, achieving an upward identity advancement. For business-side personnel, mastering data means gaining "the courage to confront" superiors (S02). Respondents noted that data empowerment has allowed them to break the bottom-tier role of "passively executing orders" and instead gain "the ability to propose solutions and even confront superiors" (S02). For data experts, this elevation of discursive authority is manifested in the transformation from auxiliary back-office support to "decision gatekeepers."

As one respondent stated, data serves as "ironclad evidence," enabling the data team to move beyond being mere advisors to becoming "decision gatekeepers" (S08). By proving logic through data, individual workplace weight is increased, and some even transform from executors into "project co-creators and operators" (S09).

Secondly, this power expansion driven by technology can generate friction with traditional experiential authority when reaching deep-water business areas, thereby exposing "experience barriers of business operations." While data can provide precise quantitative analysis, pure technological logic often fails in areas such as "commercial negotiation and complex interest coordination" (S10). Respondents generally noted that the core barrier for business experts lies in their profound understanding of the market and user pain points. Therefore, even if the data side provides analysis, if it lacks business depth, it will still be viewed as "impractical, like a castle in the air" (S11). This friction suggests to data teams that discursive authority comes not only from technological precision but also from the ability to penetrate the business context.

Following the power struggle, organizations typically establish a new order through a clear "division of final decision-making power." Interviews show that although the weight of data roles is increasing, the final "say" remains clearly demarcated. Respondents generally agreed that data experts are only responsible for "stating facts based on data," while "the ultimate decision-making power belongs to the business leader" (S01, S09, S13). Respondents used vivid metaphors to define this division: the data side is positioned as the "microscope," responsible for seeing details, while the business side is the "brain," responsible for the final decision direction (S09). In some contexts, the data team is even assigned the functions of "brakes" or a "rider," using metric red lines to constrain the blind expansion of the business side (S04). This division of power ensures that while utilizing data for efficiency, the strategic direction remains controlled by those who bear the ultimate responsibility.

Finally, this reconfiguration of power relations ultimately leads to a collaborative model of "complementarity between business and data experts." Respondents described this relationship as mutually complementary partners (S13) or a "teacher-student relationship" (S01). In this complementary logic, data experts focus on providing the "foundation"—ensuring professional data handling, tool development, and accuracy support—while business experts are responsible for "articulating core value" on this foundation, providing business understanding and analytical directions (S02, S09, S10). This collaboration with clear roles

not only reduces "ambiguous bickering" in meetings but also achieves an efficient decision-making closed-loop where "data sets the direction and experience makes the judgment."

4.5 Theme 4: Data Quality, Infrastructure, and Issues of Trust

This theme examines the organizational conditions under which BI systems either function as reliable decision-making resources or fail to do so. Rather than treating data quality as a purely technical concern, the findings suggest that trust in BI systems is produced and undermined through a layered set of organizational processes. Across the seven codes within this theme, a shared tendency can be observed: when weaknesses at the technical and foundational levels cause friction in collaboration, institutional pressure drives team members to produce performative metrics instead of real insight.

At the foundational level, the reliability of any BI system depends on the integrity of the data architecture beneath it. Respondents described metric system construction as an already-interpretive process. As S05 explained, "the drill-down dimensions are based on analysts' past successful analysis experience — dimensions where anomalies are likely to appear or where there are actionable levers — and we include them directly as BI drill-down dimensions," and noted that "generally when building dashboards, the core metrics will have two or more dashboards for cross-validation." Yet this careful groundwork is frequently disrupted by disorder in the underlying architecture. S04 described a situation in which "our company's data foundation is very chaotic — we introduced PlatformX BI in 2023 to replace old systems, but it couldn't fully replace them, so we still use both platforms today." Departmental interests further compounded this fragmentation: as S04 observed, "every person has their own table — subsidiaries have different departments, each with their own tables, product teams have product tables with fields they can't put on record but can't ignore, so there's a lot of gray area." The consequence of such disorder extends beyond inconvenience. As S05 warned, "if a company's infrastructure or underlying data has problems, the BI will be highly misleading and delay business actions." Infrastructure disorder is therefore not simply an operational difficulty but a condition that shapes the reliability of everything built on top of it.

These foundational weaknesses translate directly into difficulties at the collaboration level. Metric alignment, which involves establishing a shared definition of what a given indicator actually measures, was consistently described as one of the more contested dimensions of data work. S10 noted that "the biggest issue is metric calibration — at the data source level and for aggregate or complex calculated metrics, if there's no consensus, discussions about how to

understand a metric can go on and on," while S11 reinforced this point: "the most common issue is metric calibration — if unclear, it can go back and forth for a long time, because the same metric with different statistical scopes may yield different results." Uneven data literacy across organizational levels compounds this further. S12 described how, when a metric is statistically unreliable, "your superior often doesn't know the reason, so you spend a lot of time explaining why you can't just look at one number or metric." S13 observed more broadly that "everyone understands and interprets data differently, so there are misunderstandings, gaps, and sometimes conflicts." Rather than operating as strategic partners, data teams frequently find themselves managing the communication costs produced by the organization's uneven capacity to read and interpret data.

When these technical and collaborative difficulties remain unresolved, a third dynamic emerges at the institutional level: data activity becomes oriented toward managing appearances rather than generating genuine insight. Respondents noted that many dashboards are built but rarely consulted. S05 observed that in reality, "many analytics teams build piles of dashboards, but the download count or daily view count is nearly zero," while S01 described a related tension: "dashboard proliferation — everyone needs their own dashboard but actual usage frequency is low; the maintenance cost is very high." More significantly, respondents described instances in which data was used selectively to support predetermined conclusions rather than to inform open inquiry. S04 referenced a widely shared practice, captured in a well-known industry saying: "there's no wrong data, only wrong calibration — if you think you've crossed the red line, just change the calibration and it disappears." S13 described a parallel dynamic in which "a large portion of walk-in customers came through channel vouchers, but to hit their KPIs, agents brought in people with no real purchase intent." In some cases this extended to a broader organizational pattern. As S04 described, the underlying logic is one of deferral rather than resolution:

> We don't manage data for efficiency — just to play with numbers and maintain apparent health in front of headquarters. It's really about wrapping the bomb layer by layer and passing it down the line, making sure it doesn't explode in front of this leader or that leader. (S04)

Taken together, these findings indicate that the trustworthiness of BI systems cannot be assumed as a technical baseline. It is produced through, and frequently undermined by, the social and institutional conditions in which data work takes place. When infrastructure is stable, metric definitions are shared, and organizational incentives support transparency, BI

can serve as a reliable basis for decision-making. When these conditions are not in place, the gap between formal data governance and actual practice can widen in ways that are difficult to detect and harder still to reverse.

4.6 Theme 5: The Evolution of Job Functions and Role Transformation

The findings within this theme document how BI adoption is reshaping professional identities and job boundaries for individuals whose work involves data. Based on the codes, we can deduce that current data-related roles are diversely defined with blurred boundaries. While automation is replacing routine tasks, it is simultaneously catalyzing demand for new, hybrid roles. Furthermore, with the ongoing integration of AI, data professionals are experiencing a sense of uncertainty regarding the long-term value of the expertise they have accumulated over the years.

The starting point for understanding this transformation lies in how respondents currently define their own roles. Across interviews, these definitions varied considerably, reflecting the organizational heterogeneity in how data work is positioned and bounded. Some respondents described clearly delineated technical functions, with S15 explaining that their work involves "supporting the company's BI business development and maintenance, under the data middle platform team," while independently managing both online and offline ETL processes and providing basic data and reports to other departments. Others occupied more ambiguous middle-ground positions. S05 described a team deliberately positioned outside conventional categories, as "neither a business team nor a data team" but rather a connector serving as "a bridge between the two sides." S04, by contrast, described a role defined almost entirely by its downstream position in the information flow, with the team "passively responsible only for data that has already been recorded and acted upon by the business side." These varying self-definitions are not merely matters of job title but reflect deeper organizational uncertainties about where data work begins and ends, who legitimately owns interpretive authority, and what value data roles are expected to produce.

Against this backdrop of role ambiguity, a simultaneous restructuring is also being considered. On one side, automation and BI adoption are systematically displacing the most routine dimensions of data work. S11 observed that marginalization "mainly affects tasks that used to rely on manual operations, especially work involving the manual collection and organization of data," noting that "basic data entry or data cleaning may gradually disappear" alongside roles previously responsible for producing fixed routine reports. S15 offered a concrete

illustration of the scale of this displacement: "before, there were at least 3 people online; now it's just me and one intern, because traditional manual reports have been fully replaced and marginalized by updated tools like ITR and BI reports." On the other side, this displacement is accompanied by the emergence of new hybrid role requirements that combine analytical capability with business judgment. S14 described the direction of this shift: "analysts are no longer just doing analysis and visualization — they increasingly need to communicate with AI and train AI," while S15 noted that their organization was preparing to hire "a new role focused on data governance, compliance, and export management." Running through both of these shifts is a parallel upgrading of soft skill expectations. S11 observed that "a high-level operator basically needs to be proficient in both business and data — only then is their work more convincing and precise," and S13 articulated the deeper implication: "the focus has shifted — no longer staying on the surface of data, but deeply understanding the story behind the data, which raises the bar for analytical capability." Taken together, these accounts suggest that the transformation is not simply one of subtraction but of simultaneous displacement and recomposition, in which the declining value of mechanical execution is matched by rising demand for interpretive and communicative competence.

Underlying this restructuring is a deeper sense of professional uncertainty that several respondents expressed when reflecting on the longer-term implications of AI integration. This uncertainty is not abstract but grounded in specific observations about what AI is already capable of doing. S06 noted that "if AI integration can highly efficiently optimize BI data and calculations, I believe a lot of manpower can be cut," while S14 described a more personal dimension of this concern: "all professional knowledge and experience needs to be fed into AI — that gives me some anxiety. After an analyst's experience is given to AI, will the analyst's value diminish? " This question captures something that goes beyond the standard discourse of automation and job displacement. It points to a particular kind of professional vulnerability: the possibility that the very act of contributing expertise to AI systems may accelerate the erosion of the value that expertise once conferred. At the same time, not all respondents shared this sense of threat. S07 suggested that AI currently lacks the capacity to address management needs in ways that make human analysts indispensable in the short term, while S10 observed that AI could help analysts access frontier methods and lower technical barriers. These divergent responses suggest that the experience of professional uncertainty under AI integration is itself unevenly distributed, shaped by individual role positioning, organizational context, and proximity to the capabilities that AI is most rapidly acquiring.

The picture that emerges across these accounts is one of a professional field in transition, where the pressures of change are real but their ultimate direction remains open. The dissolution of routine roles, the emergence of hybrid competency demands, and the advancing capabilities of AI are not experienced uniformly but filtered through individual circumstances, organizational histories, and the particular ways in which different roles engage with data. What cuts across these differences, however, is a shared recognition that technical skill alone is no longer sufficient, and that the meaning of data work itself is being renegotiated in ways that have yet to fully settle.

5. Discussion

5.1 BI as Sociomaterial Practice

The empirical findings of this study support and further develop the concept of constitutive entanglement proposed by Orlikowski and Scott (2008). In traditional organizational management perspectives, BI is often understood as a neutral and external technological tool. However, the descriptions from participants in Theme 1 suggest that the boundary between technology and organizational practice is not so clear.

This entanglement can be seen most clearly in the repeated mention of stoppage panic (Code 04: Anxiety about losing BI support) among respondents. When BI systems become the “retina” of the organization, they are no longer optional support tools. Instead, they become embedded in how organizational reality is constructed and experienced. As one participant noted: "If BI is stopped, daily operation and supervision will directly face paralysis" (S09). This sense of operational paralysis shows the materiality of BI. The real-time flow of data and visual interfaces shape employees’ daily routines in ways that are difficult to separate from the work itself. This suggests that BI practice should not be understood only as humans using tools. It should be understood as a sociomaterial assemblage, in which technological affordances and human agency are closely intertwined.

However, the findings of this study also complicate some assumptions in sociomateriality theory about technological stability. Findings from Theme 4 show that this entanglement is not a one-way process of technological empowerment. It is also shaped by social forces. Data formalism (Code 26: Formalism and data decoupling) and risk-evasive data behaviors (Code 27: Risk avoidance and transfer of responsibility) show that internal power relations and appraisal pressures can limit the material effects of the technology. When BI is used as a

defensive strategy to improve the appearance of performance or avoid accountability, its original focus on objective accuracy is shaped by a social logic of survival.

This finding suggests a need to refine Orlikowski ' s view of entanglement. Sociomaterial entanglement should not be understood as a smooth process of fusion. Instead, the evidence in this study shows that it is more dynamic and contested. It also involves continuous political negotiation. The form and degree of this entanglement may differ across organizational roles. For top management, BI may act as strategic scaffolding for decision-making. For frontline employees, it may function more like materialized algorithmic control. These observations suggest that studies of BI practices should pay more attention to how the material properties of technology, such as data granularity, and specific social institutional environments, such as KPI culture, shape each other through mutual constitution.

5.2 BI and Organizational Sensemaking

Examined through Weick ' s (1995) framework of organizational sensemaking, the introduction of BI systems does more than accelerate information flow. It also reshapes how members extract cues from ambiguous business signals and give them meaning. In Theme 2 of this study, data becomes a central logic within the decision chain. However, the empirical results suggest that BI does not simply remove uncertainty by providing truth. Instead, it functions as a contested space within the sensemaking process.

The data from this study partly supports Namvar et al.'s (2016) argument that BI can provide scaffolding for sensemaking. In a data-driven culture (Code 14: Data-driven business logic), metrics become a shared language for internal organizational communication. However, this supporting role also comes with clear tensions. One important finding is the game between experience and data (Code 11: Experience, intuition, and subjective judgment; Code 12: Data-based validation and the overturning of prior assumptions), which connects with Weick ' s distinction between plausibility and accuracy.

When the outputs from BI dashboards conflict with a manager ' s intuition, organizational members may give more weight to the plausibility of existing narratives than to the direct acceptance of data accuracy. As one respondent observed, when data conflicts with established judgment, the business side tends to prioritize questioning "whether the data parameters and experimental design are flawed" (S12). This example suggests that the cues

provided by BI can only become a basis for action after they are examined through experience and connected to an understandable narrative framework.

This study also provides some micro-level observations that may refine the view of Namvar et al. (2016) BI can provide decision support, but it may also lead to reductive sensemaking in some contexts. Respondents noted that in environments with strong reliance on metrics, people may overlook the social complexity behind the numbers, criticizing BI's inherent bias toward presenting aggregated averages that makes it "difficult to reflect the localized process value" (S09). This suggests that BI practices may simplify rich organizational realities into abstract figures. As a result, they may limit members' ability to understand context-specific details. Therefore, the role of BI in organizations is not only to support decision-making. It also works as a cognitive filter. It increases the visibility of some dimensions, but it may also create new interpretive blind spots.

5.3 BI Practices and Role Reconfiguration

The findings across Themes 3 and 5 are closely related to the theoretical discussion of role reconfiguration developed in Chapter 2. Bosworth and Kreps (1986) describe traditional organizational configurations as based on stable role-playing. In this situation, actors form their behaviors and expectations through established rules and a stable status-role nexus. The empirical evidence in this study suggests that BI adoption creates a continuous disruption to this stability. It pushes organizational members toward role-making. This is a more flexible process in which roles are actively negotiated in response to changing conditions, rather than simply inherited from existing scripts. Two mechanisms seem to drive this process. The first is the increased behavioral visibility created by BI systems (Leonardi & Treem, 2020). The second is the selective displacement of some types of work through algorithmic control (Schafheitle et al., 2020).

At the organizational level, behavioral visibility changes the boundaries of interpretive authority. This can be seen clearly in Theme 3. As data becomes more accessible, actors who can connect technical logic with business context gain new forms of standing in the organization (Code 16: Voice advancement and role progression). S01 described this shift directly: before, when data was opaque, our role was more like a data-fetching tool. Now with BI, the business side can see 80% of daily data themselves, and our role has become explaining, judging, and advising. However, this redistribution is still contested. The

allocation of final decision-making authority continues to be negotiated rather than settled (Code 15: The allocation of final decision-making authority). Respondents often distinguish between those who provide data and those who have the right to act on it. This tension is also shaped by structural gaps between management and frontline execution teams (Code 20: Differences in perspective between management and operation teams).

At the professional level, the pattern shows both displacement and recomposition. BI adoption reduces the importance of routine data work. At the same time, it creates demand for hybrid roles that combine analytical capability with business judgment (Code 29: The marginalization and replacement of basic work; Code 30: The emergence of demand for hybrid roles). S15 described this displacement clearly: "before, there were at least 3 people online; now it is just me and one intern, because traditional manual reports have been fully replaced." This does not only reduce discretion, as Schafheitle et al. (2020) suggest. It also redirects discretion from mechanical execution toward interpretive and communicative work (Code 31: Upgrading soft skills and communication ability).

Across these two levels, the findings show a broader redistribution of interpretive authority. Ranson et al. (1980) describe this redistribution as central to organizational structure. In this study, this redistribution is taking place, but it is uneven. It is shaped by the organizational contexts and industry conditions in which different actors encounter BI. Plesner and Justesen (2025) note that the process of becoming algorithm-ready differs across organizational contexts and is rarely simple. The variation among the fifteen respondents in this study also supports this point. Some respondents had already moved from executor to strategic partner. Others still had data work that was mainly understood as a support function with limited interpretive standing (Code 28: Job responsibilities and boundary definition). This variation suggests that role reconfiguration under BI adoption should not be understood as a uniform transformation. It is better understood as a set of ongoing and context-sensitive negotiations, with outcomes that remain open.

5.4 Theoretical Contributions

This study explored how BI practices reconfigure organizational dynamics across different industries and organizational roles. It engages with three bodies of literature: sociomateriality, organizational sensemaking, and role dynamics. These three areas have often developed separately, rather than in close connection with each other. One contribution of this study is to

bring these three perspectives together in one empirical study. Existing BI research has often focused on technical performance or on separate dimensions of organizational behavior. The findings of this study demonstrate that a more integrated perspective can help explain why the same BI tools produce different outcomes across different organizational settings. This point is consistent with the gap identified by Mekimah et al. (2024), who note that the integration of BI with organizational decision-making processes remains underexplored in the literature.

A second contribution of this study concerns data formalism. In this study, data formalism refers to the systematic use of data to manage appearances rather than to generate genuine insight. Much of the existing literature on BI and organizational sensemaking assumes that data will be used in good faith once it becomes available. The findings in Theme 4 complicate this assumption. Respondents described selective metric calibration, deliberate manipulation of performance indicators, and organizational practices that defer problems rather than solve them. These patterns seem to be shaped less by individual misconduct and more by institutional pressures and incentive structures in data work. Consequently, the relationship between data accessibility and organizational learning is far more contingent and politically contested than current theoretical accounts acknowledge, underscoring the need for theoretical frameworks to internalize the structural conditions under which BI systems generate institutional trust or systemic skepticism.

5.5 Practical Implications

The empirical results of this study have theoretical significance. They also offer practical implications for organizations that are dealing with digital transformation, especially in BI implementation and talent management.

For data governance, the evidence reveals that metric standardization and data literacy training should not be understood only as technical tasks for the IT department. They can also be understood as ways to rebuild organizational trust over time. Based on the findings in Theme 4, data conflicts caused by the lack of unified standards can create friction and mutual distrust in cross-departmental collaboration. For this reason, organizations may need to build cross-functional communication mechanisms. These mechanisms can help address technical conflicts through more formal organizational processes. Instead of focusing only on data cleaning, organizations may also need to align the underlying logic behind data use. This shift

in governance orientation may help reduce the internal frictional costs described by respondents.

Regarding role design and collaborative workflows, the findings show the potential value of more equal collaboration between data and business stakeholders. Theme 3 showed a discourse power imbalance in BI practices. In some cases, this imbalance created a rigid division of work, where business teams proposed requirements and data teams produced reports. Embedded analytical structures may help address this problem. In these structures, data experts participate more directly in business decision-making processes. This may help reduce the cognitive gap between technical and business perspectives. Drawing on the patterns observed across Theme 5, data roles may become more productive when they shift from back-end supporters to front-end enablers, although the conditions for this transition will differ across organizational contexts.

For AI-driven automation, the findings in Theme 5 showed career anxiety among junior employees. They also showed a growing organizational demand for hybrid competencies. Organizations facing these pressures may need to focus not only on workforce reduction, but also on supporting role transitions. This means helping data practitioners develop broader business understanding together with technical skills. Building a culture of continuous learning may help organizations respond to the changing role demands created by the convergence of BI and AI. However, the specific form of this response will depend on the industry and organizational context.

5.6 Limitations and Future Research

Although this study provides profound empirical insights into how BI practices reconfigure organizational dynamics, certain limitations remain, presenting broad opportunities for future exploration.

The first is sample limitations. The 15 participants in this study are highly concentrated within major tech firms in the Chinese internet sector. These medium-to-large enterprises generally possess a very high degree of digital maturity. Consequently, the generalizability of these findings, particularly regarding complex data games and deep power restructuring, may be limited when applied to smaller enterprises or non-Internet industries that are still in the early stages of digital transformation.

The second involves methodological limitations. This study relies primarily on deep interviews conducted at a single point in time, essentially making it a cross-sectional study. Due to the lack of longitudinal data, the research captures changes primarily through participants' retrospective accounts and is unable to track the dynamic evolutionary process of role reconfiguration and shifts in power relations in real time.

Finally, there are contextual limitations. This study is deeply rooted in a specific Chinese organizational culture. The recurring themes of risk aversion and data formalism observed in the interviews are largely shaped by the distinct hierarchical notions and rigorous KPI culture unique to the Chinese internet industry. Such highly contextualized findings might affect the cross-cultural applicability of the established theories.

Building on these limitations, future research could proceed in the following three directions. First, conduct cross-cultural comparative studies. Future scholars are encouraged to replicate this study within Western organizational contexts, such as European companies that emphasize flat structures and employee empowerment, to test and compare the universality and heterogeneity of sociomateriality and sensemaking mechanisms across different cultural landscapes. Second, adopt longitudinal studies designs. Future research could select an organization that is preparing to implement a BI system and conduct long-term field observations. This would allow researchers to capture the dynamic evolution of role reconfiguration before, during, and after the technology's implementation in real time, thereby illustrating the mutual shaping of technology and organization more delicately. Third, advance quantitative verification. Researchers could develop standardized measurement scales based on the qualitative themes extracted in this study, such as measuring the degree of tension between experience and data, or the tendency toward data formalism. By conducting hypothesis testing across a larger and more diverse industry sample, scholars can further consolidate and extend the theoretical framework proposed in this research.

6. Conclusion

This study examined how Business Intelligence (BI) practices reconfigure organizational dynamics through sensemaking and role reconfiguration within digitally mature sectors. It draws on 15 semi-structured interviews with data practitioners and business managers in China. The analysis uses three theoretical perspectives: sociomateriality, organizational sensemaking, and role dynamics. The findings challenge the common technocentric view of

BI. BI adoption is not only a smooth and rational process. It is also a complex, contested, and social process.

Across the five empirical themes, a clear and definitive pattern emerges: BI does not simply improve decision-making by passively making more data available. Instead, it actively intervenes in the process of constructing organizational reality. Specifically, we draw three conclusions regarding how BI reconfigures organizational dynamics:

First, in terms of sensemaking, BI generates profound friction between data logic and experiential judgment. Rather than instantly resolving ambiguity, BI dashboards become contested spaces. Employees do not blindly follow data; instead, they constantly negotiate and battle between algorithmic evidence and their historical business intuition.

Second, BI leads to role reconfiguration and changes traditional hierarchies. By increasing behavioral visibility, BI redistributes interpretive authority. It moves data practitioners from passive back-office supporters to more active participants in decision-making. At the same time, it places traditional business managers under new forms of algorithmic control. This change creates a continuous negotiation over who can be seen as a legitimate expert.

In the end, friction in how teams make sense of data, combined with shifting power dynamics during role reconfigurations, ultimately turns decision-making into a highly political process. Within this context, the promise of objective transparency is frequently undermined by institutional survival instincts. The tension between formal data governance and actual practice often breeds "data formalism." When faced with rigid performance pressures, actors tend to manipulate or selectively interpret metrics to deflect risk and maintain appearances, proving that data is easily decoupled from genuine business insights.

In answering the research question, these findings demonstrate that BI is not a neutral mirror reflecting organizational reality, but a disruptive social force that actively reshapes it. Its organizational consequences are deeply molded by the power struggles and institutional incentives into which it is introduced. To truly leverage BI, organizations must move beyond the myth of technological neutrality and confront the underlying political and social dynamics of data work.

References

Abusweilem, M., & Abualoush, S. (2019). The impact of knowledge management process and business intelligence on organizational performance. *Management science letters*, *9*(12), 2143-2156. https://doi.org/10.5267/j.msl.2019.6.020

Bačić, D., & Fadlalla, A. (2016). Business information visualization intellectual contributions: An integrative framework of visualization capabilities and dimensions of visual intelligence. *Decision Support Systems*, *89*, 77-86.

Boland Jr, R. J. (1984). Sense-making of accounting data as a technique of organizational diagnosis. *Management Science*, 30(7), 868-882.

Bosworth, S. L., & Kreps, G. A. (1986). Structure as Process: Organization and Role. American Sociological Review, 51(5), 699–716. https://doi.org/10.2307/2095494

Božič, K., & Dimovski, V. (2019). Business intelligence and analytics for value creation: The role of absorptive capacity. *International journal of information management*, *46*, 93-103.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative research in psychology, 3(2), 77-101.

Bryman, A. (2016). Social research methods (5th ed.). *Oxford University Press*.

Chen, Y., & Lin, Z. (2021). Business intelligence capabilities and firm performance: A study in China. *International journal of information management*, *57*, 102232.

Creswell, J. W., & Creswell, J. D. (2018). Research design: Qualitative, quantitative, and mixed methods approaches (5th ed.). *Sage Publications*.

Davenport, T. H. (2010). Business Intelligence and Organizational Decisions. *International Journal of Business Intelligence Research (IJBIR)*, *1*(1), 1-12.

Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of management review*, *14*(4), 532-550.

Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases: Opportunities and challenges. *Academy of management journal*, *50*(1), 25-32.

Fink, L., Yogev, N., & Even, A. (2017). Business intelligence and organizational learning: An empirical investigation of value creation processes. *Information & management*, *54*(1), 38-56.

Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational research methods*, *16*(1), 15-31.

Jordan, J., & Ellen, C. (2009). Business need, data and business intelligence. *Journal of Digital Asset Management*, *5*(1), 10-20.

Kaplan, S. (2008). Framing contests: Strategy making under uncertainty. *Organization science*, *19*(5), 729-752.

Leonardi, P. M., & Treem, J. W. (2020). Behavioral visibility: A new paradigm for organization studies in the age of digitization, digitalization, and datafication. *Organization studies*, *41*(12), 1601-1625.

Leonardi, P. M. (2011). When flexible routines meet flexible technologies: Affordance, constraint, and the imbrication of human and material agencies1. *MIS quarterly*, *35*(1), 147-167.

Maitlis, S., & Lawrence, T. B. (2007). Triggers and enablers of sensegiving in organizations. *Academy of management Journal*, *50*(1), 57-84.

Mekimah, S., Zighed, R., Mili, K., & Bengana, I. (2024). Business intelligence in organizational decision-making: a bibliometric analysis of research trends and gaps (2014–2024). *Discover Sustainability*, *5*(1), 532.

Meyer, J. W., & Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. *American journal of sociology*, *83*(2), 340-363.

Namvar, M., Cybulski, J. L., & Perera, L. (2016). Using Business Intelligence to Support the Process of Organizational Sensemaking. Communications of the Association for Information Systems, 38, 330–352. https://doi.org/10.17705/1CAIS.03820

Olszak, C. M. (2014). Business intelligence and analytics in organizations. In *Advances in ICT for Business, Industry and Public Sector* (pp. 89-109). Cham: Springer International Publishing.

Olszak, C. M. (2016). Toward better understanding and use of business intelligence in organizations. Information systems management, 33(2), 105-123.

Olszak, C. M. (2022). Business intelligence systems for innovative development of organizations. *Procedia Computer Science*, *207*, 1754-1762.

Orlikowski, W. J. (1992). The duality of technology: Rethinking the concept of technology in organizations. *Organization science*, *3*(3), 398-427.

Orlikowski, W. J. (2000). Using technology and constituting structures: A practice lens for studying technology in organizations. *Organization science*, *11*(4), 404-428.

Orlikowski, W. J., & Gash, D. C. (1994). Technological frames: making sense of information technology in organizations. *ACM Transactions on Information Systems (TOIS)*, *12*(2), 174-207.

Orlikowski, W. J., & Scott, S. V. (2008). sociomateriality: challenging the separation of technology, work and organization. *Academy of Management annals*, *2*(1), 433-474.

Plesner, U., & Justesen, L. (2025). Making Organizations Algorithm-Ready: Algorithmic Organizing Through Techno-Organizational Scripts. In V. L.

Glaser, C. Moser, D. A. Anderson, & P. D. Jennings (Eds.), Algorithmic Organizing (pp. 69-87). Emerald Group Publishing. https://doi.org/10.1108/S0733-558X20250000095004

Ransbotham, S., & Kiron, D. (2017). Analytics as a source of business innovation. MIT Sloan Management Review, 58(3). https://sloanreview.mit.edu/projects/analytics-as-a-source-of-business-innovation/

Ranson, S., Hinings, B., & Greenwood, R. (1980). The Structuring of Organizational Structures. Administrative Science Quarterly, 25(1), 1–17. https://doi.org/10.2307/2392223

Sabherwal, R., & Becerra-Fernandez, I. (2011). *Business intelligence: Practices, technologies, and management*. John Wiley & Sons.

Schafheitle, S., Weibel, A., Ebert, I., Kasper, G., Schank, C., & Leicht-Deobald, U. (2020). No stone left unturned? Toward a framework for the impact of datafication technologies on organizational control. *Academy of Management Discoveries*, *6*(3), 455-487. https://doi.org/10.5465/amd.2019.0002

Shollo, A. (2013). The role of business intelligence in organizational decision-making. Frederiksberg: Copenhagen Business School (CBS).

Suša Vugec, D., Bosilj Vukšić, V., Pejić Bach, M., Jaklič, J., & Indihar Štemberger, M. (2020). Business intelligence and organizational performance: The role of alignment with business process management. *Business process management journal*, *26*(6), 1709-1730. https://doi.org/10.1108/BPMJ-08-2019-0342

Weick, K. E. (1995). *Sensemaking in organizations*. Sage.

Yin, R. K. (2018). *Case study research and applications* (Vol. 6). Thousand Oaks, Sage.

Appendix

1. Semi-Structured Interview Outline

Phase 1: Icebreaker & Professional Background

Q1: Could you briefly introduce your role and what you do on a daily basis? How is your team positioned? Is it a business team or a data team? - Team Member

Q1: What kind of team do you lead? Who are the members, and what are their specific responsibilities? -Manager

Q2: Do you use data or data dashboards (BI systems) in your work? Under what circumstances do you usually use them?

- Follow-up: Based on your previous work experience, is there a noticeable difference in how BI systems are used here compared to before?

Q3: When making decisions, what data do you usually rely on? Or, how does your team typically make decisions based on data or experience?

Phase 2: Daily Collaboration & Role Division

Q4: In a typical project, what kind of discussions do you have within your team or with other teams based on the data provided by BI?

- Follow-up: How do the business side and the data side usually interact? Who initiates the requirement, who provides the support, and who makes the final decision?
- Follow-up: What role do you think BI plays in your team?

Q5: From your own role's perspective, have you encountered any issues when communicating with team members or superiors due to the use of BI?

Phase 3: Conflicts, Tensions & Decision Making

Q6: Do you feel that BI systems have both pros and cons for your or your team's judgments and decisions?

Q7: Can you recall a case where your analysis influenced a business decision? What exactly happened then? - Team Member

Q8: Could you share a specific example: have you ever experienced a situation where the data results were inconsistent with your own judgment or experience?

- Follow-up: How did you or your team handle this situation?
- Follow-up: What views did different roles (like the analyst vs. the business leader) have? Whose explanation eventually convinced everyone?

Phase 4: Deep Feelings & Identity

Q9: With the current BI system, do you feel your role in the team's decision-making chain has become more important or less important? Why?

Q10: If the BI system were suddenly shut down for a week, what changes would happen to your work? - Team Member

Q10: If the BI system were suddenly shut down for a week, what changes do you think would happen to the team's decision-making or workflows? - Manager

Q11: What do you think the relationship between a data expert and a business expert should be like?

Q12: As BI tools develop, how has your insight into actual business operations changed?

- Follow-up: What impact has this had on project implementation?
- Follow-up: Has it affected how you perceive your role within the organization or team?

Q13: Has BI marginalized certain roles or created any new ones?

2. Informed Consent Form

Research Project: How do BI practices reconfigure organizational dynamics across different industries and organizational roles?

Researchers: Qiuchi Wang & Ziling Huang, Department of Business Studies,
Uppsala University

1). Research Purpose and Description

The aim of this research is to explore how Business Intelligence (BI) systems and data-driven practices influence daily decision-making processes, organizational sensemaking, and the reconfiguration of professional roles within platform organizations. You are being invited to participate because you have relevant experiences.

2). Procedures

- Participation involves a semi-structured interview lasting approximately 40 minutes.
- With your consent, the interview will be audio-recorded and subsequently transcribed verbatim.
- As the study is conducted in English, but interviews may be held in Mandarin, the transcripts will be translated for analytical purposes.

3). Confidentiality and Anonymity

- Confidentiality: All primary data, including recordings and original transcripts, will be kept strictly confidential and stored securely.
- Anonymity: To protect your identity and mitigate professional risks, your name and the specific identity of your organization will be fully anonymized in any final thesis, publications, or presentations.

4). Voluntary Participation and Right to Withdraw

Your participation is entirely voluntary. You have the right to refuse to answer any specific question and may withdraw from the study at any point without any consequences.

5). Contact Information

If you have any questions or require further information about this study, please contact the researchers:

- Qiuchi Wang
- Ziling Huang

6). Participant's Statement

I confirm that I have read and understood the information for the above study. I have had the opportunity to ask questions, and these have been answered to my satisfaction. By agreeing to take part in this study, I consent to the terms described above.

3. Table

a. Code Frame

Examples of Data Excerpts	Description	Code	Theme
This also includes reviewing BI-based data analysis. In some cases, we also build BI dashboards to continuously monitor the operational performance of a product.	Daily work involves reviewing data analysis results and personally building BI dashboards.	Code 01: Routine monitoring of core metrics	T1: Perceptions of the value of BI tools and the boundaries of their limitations
The first and most direct benefit is efficiency. Without BI, someone would have to extract the same data every day, which consumes manpower and creates repetitive work.	The BI system reduces repetitive work and improves efficiency.	Code 02: Reducing labor costs and improving analytical efficiency	
Data dependence can easily become excessive. In some cases, unexpected market events may occur, and some users may still feel that the actual experience is not good enough. If such data are shown to users, they often cannot tell what has been changed. When one looks only at the data, the actual adjustments that have been made tend to be overlooked. Therefore, I believe that highly personalized and finely segmented businesses are generally not well covered.	Excessive reliance on data may create blind spots, because data cannot fully capture sudden market changes or the real experiences of users in highly segmented and personalized scenarios.	Code 03: Data blind spots and limitations of prediction	

Without it, I cannot assess whether a given solution works effectively. I can only rely on my own experience and intuition. When problems arise, I am also unable to make timely adjustments, which can easily lead to subsequent losses. In that case, how should compensation be handled?	Without the real-time monitoring capability provided by the BI system, work would rely largely on intuition, which increases the risk of delayed response and business losses.	Code 04: Anxiety about losing BI support	
From the revenue data, revenue remains relatively stable during the period when the negative review rate declines, but the probability of drivers being penalized increases.	BI helps identify specific business trends within a given period.	Code 05: Detecting irregularities and identifying problems	T2: The role and logic of data in the decision-making chain
For example, if metrics such as GMV or GTV decline, further analysis is required. This analysis should examine the transaction chain, the consumer side, the business side, and the platform as a whole to determine which part drives the change. It should also assess the situation from both internal and external perspectives, such as whether competitors are affected as well.	When core indicators show abnormal declines, the issue must be investigated from multiple perspectives, including different business endpoints and competitor dynamics.	Code 06: Attributing fluctuations and breaking down the operational chain	
Later, I looked back at the BI system and pulled the channel data for the most recent three months. The data at that time show that its customer acquisition cost is already much lower than that of other channels, and user retention afterward is also relatively low. In contrast, its conversion performance is clearly better. The only issue is that our company had not paid enough attention to it before.	By examining historical BI data, one may identify a low-cost, high-conversion channel that had previously been overlooked by the company.	Code 07: Deeper business insight and analytical logic	
I find that for the same merchant on Meituan, the meal price is higher than that of a competitor I see on Douyin. In some cases, the competitor also offers set meals while I do not. This type of issue can also be reflected in the BI system. When such a situation has a substantial impact on transaction volume, if I am in a sales role, I will follow up with the merchant directly.	Frontline sales staff can use the BI system to detect price differences relative to competitors and immediately translate these findings into merchant follow-up actions.	Code 08: Competitor and external environment analysis	
We use the BI system to extract stratified user data and examine conversion, retention, and risk levels across different customer segments,	BI supports the formulation of precise targeting strategies that balance growth	Code 09: Decision support and analytical justification	

which helps define a precise targeting strategy for each group. This approach ensures growth performance while preventing high-risk users from being introduced.	objectives with risk control.		
By comparison, we find that this random bidding strategy, in which the traffic of the entire group is adjusted uniformly to 1.1 times or 0.9 times at random, yields higher returns than the model-based bidding strategy. This indicates that the logic of the model itself is flawed.	An A/B test shows that random bidding yields higher returns than algorithmic bidding, which invalidates the logic of the model.	Code 10: Strategy iteration and A/B test	
So, in essence, data mainly help us make decisions more efficiently, but they do not necessarily make our decisions better. The quality of a decision still depends more on people making judgments based on their own business experience.	Data can make decision-making more efficient, but high-quality decisions still depend on business experience.	Code 11: Experience, intuition, and subjective judgment	
Intuitively and subjectively, one may assume that a large proportion of older adults should use the large-font version. However, once I check the data, I find that the proportion is actually not high.	Subjective assumptions about the needs of elderly users may differ substantially from actual operational data.	Code 12: Data-based validation and the overturning of prior assumptions	
In other words, it must be closely aligned with the business and match actual business needs. Otherwise, no matter how polished or sophisticated the dashboard looks, nobody will use it if it does not fit the business context.	BI dashboard design must align with actual business needs; otherwise, even a technically sophisticated system will be abandoned.	Code 13: Data detached from the real business context	
People will review these data very frequently. After doing so repeatedly, they tend to check the data before making decisions and habitually conduct some preliminary analysis in advance. Over time, this process helps them develop data literacy and strengthens their ability to analyze data.	BI gradually cultivates a data-driven mindset among business teams by encouraging them to consult data before making decisions.	Code 14: Data-driven business logic	
In fact, I believe the business team is mainly responsible for raising requirements and making decisions, while the data team serves a supporting role and mainly needs to ensure that the data are highly accurate.	The business side is responsible for raising requirements and making decisions, whereas the data side mainly provides accurate data support.	Code 15: The allocation of final decision-making authority	T3: Power relations and discursive authority associated with data roles

When data were not transparent in the past, our role was more like a data extraction tool. Whoever needed data would come to us, and we would simply pull it for them. However, after BI became widely adopted, the business side could access about 80% of the routine data on their own. As a result, our role instead shifted toward explaining the data, diagnosing problems, and offering recommendations, that is, serving as a data advisor.	As data accessibility improves, the role shifts from passively extracting data to proactively offering diagnosis and recommendations.	Code 16: Voice Advancement and Role Progression	
The two sides reinforce each other. They have a partnership rather than a relationship in which one dominates the other. Instead, they complement and support each other.	Data experts and business experts should be regarded as complementary partners rather than competing actors.	Code 17: Complementarity between business experts and data experts	
If you have done it yourself and have achieved successful results, your analysis becomes far more convincing. You can demonstrate not only that you perform better in practice, but also that you bring strong professional expertise in data analysis. In that case, others are naturally more willing to accept your judgment. This also suggests that, if one wants to develop a career in data analysis, it is essential to understand the business, and it is even better to have gained hands-on business experience for some period of time.	For long-term career development, data analysts need not only analytical skills but also practical business experience and proven success in the field.	Code 18: The barrier created by hands-on business experience	
When I ask the data team to build it, the business side is relatively small, while the data team is a larger, independent function. Their perspectives are therefore different. The data team may feel that this kind of work is difficult to implement, which can then lead to some conflicts over project progress.	Conflicts over project progress often emerge because business teams and data teams assess implementation difficulty and workload from different perspectives when building dashboards.	Code 19: Cross-department communication and collaboration frictions	
The metrics that senior management focuses on may not be that comprehensive, but they are usually more central and target the key issues. By contrast, as the person responsible for a specific module, you are more	Senior management tends to focus on a few core indicators, whereas module owners are more concerned with the full chain of related data.	Code 20: Differences in perspective between management and operation teams	

likely to examine the data in a more complete way, including its background, development, and performance across different stages of the chain.			
Tools such as SQL are more like low-level instruments, somewhat similar to working directly with code. They can be used for many purposes and are generally suited to basic analysis, data analysis, or tasks such as data extraction. By contrast, BI is more appropriate for long-term monitoring and for supporting management with visualized reports, as well as helping them make decisions.	SQL and similar coding tools provide flexibility for low-level and foundational analysis, whereas BI is more suitable for long-term monitoring and visual reporting for management.	Code 21: Data tools and metric system construction	T4: Data quality, infrastructure, and issues of trust
If a company's infrastructure, especially its underlying data, has serious problems, as in one company I joined later, then its BI system can become highly misleading and may delay business actions.	If the underlying data infrastructure is flawed, the data presented in BI dashboards can be highly misleading.	Code 22: Base data governance and disorder	
Another issue is the proliferation of dashboards. Everyone wants to have their own dashboard, but in practice, the actual frequency of use is not very high. As a result, the maintenance cost becomes quite high.	The widespread creation of personal dashboards creates a tension between high maintenance costs and low usage frequency.	Code 23: Resource redundancy and system overhead	
Different source paths also create problems. As a result, even when some data come from the same source, their definitions, metrics, and levels of granularity may still differ. In particular, when the granularity is too broad or too coarse, the overall cost of data consolidation and organization increases.	Inconsistent data sources, inconsistent definitions, and overly coarse granularity all increase the cost of data consolidation.	Code 24: Difficulties in metric alignment and consensus building	
Another issue arises in cross-department communication. Colleagues from other teams are often not very familiar with BI dashboards. As a result, when I use these data to communicate with them, I need to spend additional time explaining what the indicators mean and how the data are generated. Otherwise, people cannot fully understand the information, which makes it harder for us to communicate effectively and reach a shared understanding.	In cross-department communication, unfamiliarity with the BI system on the part of other teams creates additional explanation costs and communication barriers.	Code 25: The cost of data explanation and user education	

Because it is mainly used for upward reporting, I think it becomes somewhat formalistic in this context. Moreover, some metric definitions can even be adjusted. In such cases, the data reflected in the BI system are not necessarily objective and may instead mislead management.	Because BI is often used for upward reporting, it may become formalistic, and adjustable metric definitions may distort the data and even mislead management.	Code 26: Formalism and data decoupling	
If the issue remains unresolved for a long time, we may need to issue a penalty notice or escalate the matter through further communication and other forms of cross-department coordination.	When abnormal data are not addressed in time, cross-department pressure may be applied.	Code 27: Risk avoidance and transfer of responsibility	
Our team is generally positioned in a dual-facing and dual-functional role. It is neither a business team nor a data team. Instead, it can be understood as a connector that serves as a bridge between the two sides.	The team is positioned as a data-business unit that serves as a bridge between the two sides.	Code 28: Job responsibilities and boundary definition	T5: The evolution of job functions and role transformation
Marginalization mainly affects tasks that used to rely on manual operations, especially work involving the manual collection and organization of data. For example, basic data entry or data cleaning may gradually disappear. Similarly, some roles that were previously responsible for producing fixed routine reports may also become obsolete.	Basic data entry, manual data organization, and routine report production are being marginalized by automation.	Code 29: The marginalization and replacement of basic work	
It also requires you to understand the data and know how to use them in practice. What it demands is a more well-rounded and versatile type of talent.	Business roles require not only execution but also the ability to understand and apply data effectively.	Code 30: The emergence of demand for hybrid roles	
The focus may shift. It is no longer enough to stay at the surface level of the data. Instead, one needs to develop a deeper understanding of the story behind the data, which places higher demands on technical skills and analytical ability.	The emphasis of many roles is shifting toward a deeper understanding of the story behind the data, which raises the standard for analysis.	Code 31: Upgrading soft skills and communication ability	
If the integration of AI can substantially improve the efficiency of BI data optimization and data processing, then I do think that a considerable amount of human labor can be reduced.	Future AI integration into BI is likely to lead to a substantial reduction in labor demand.	Code 32: Career anxiety in the face of AI substitution	

b. Code Book

Code ID	Code	Definition	Quote Examples
Code 01	Routine monitoring of core metrics	Respondents make viewing BI dashboards a fixed daily habit, continuously monitoring core business metrics to assess operational health.	2-4: "I usually look at user-related data, new user count, retention rate, to judge user quality and check the final conversion rate." 6-15: "From the GMV metric we can clearly see that the scale indicator has declined."
Code 02	Reducing labor costs and improving analytical efficiency	Business teams use BI tools to pull data independently, eliminating reliance on technical schedules and analysts.	1-4: "My current company has much finer data granularity and stronger real-time capability, everyone can pull their own data without depending on analysts." 8-10: "Before, the data team supported the business side and we were passive recipients waiting for reports. Now we are both business and data producers and consumers, so BI has become our daily tool."
Code 03	Data blind spots and limitations of prediction	Respondents recognize structural limitations of BI tools in data coverage and predictive capability.	1-18: "We may fall into data superstition, we cannot quantify certain factors, like a brand's long-term impact, which cannot be seen in just one week or one month." 9-12: "BI systems naturally lean toward presenting aggregate averages and find it hard to capture partial structures or relatively complete process details." 9-14: "Our work involves innovation, cross-category efforts, and non-standardized metrics, sometimes the existing data framework simply cannot compute them."
Code 04	Anxiety about losing BI support	Strong sense of dependence and functional anxiety expressed when respondents were asked to imagine losing the BI system.	9-24: "Daily operational oversight would simply come to a halt." 9-25: "We would have no idea what is happening in the market, and risk warnings would also fail."

Code 05	Detecting irregularities and identifying problems	Respondents use BI dashboards or data reports to identify abnormal fluctuations in business metrics, triggering further investigation and attribution analysis.	3-6: "If a metric for Department A drops the most and fastest, or is simply an anomalous indicator, it automatically appears in the top-right corner as a special situation for this week or month, and the system automatically performs some analysis." 12-32: "Using BI to more quickly identify which user group or which dimension is driving growth or causing the anomaly, the progress in BI allows us to understand the business more precisely."
Code 06	Attributing fluctuations and breaking down the operational chain	After detecting a metric anomaly, respondents conduct systematic multi-dimensional attribution analysis along the business chain.	5-15: "If GMV/GTV drops, I need to analyze it, from the transaction chain, from C-side and B-side, from the platform as a whole, from competitors inside and outside the industry." 5-26: "When an anomaly is detected, only the analyst can trace back why this endpoint has such a big problem, then investigate what happened with the deployment team."
Code 07	Deeper business insight and analytical logic	Respondents describe a cognitive upgrade from surface results to deep causal understanding of business logic through BI tools.	2-24: "Before, I could only see surface results — sales today, visitors today — but not why. I relied on personal experience and intuition. With the BA system, I can see the entire business chain in detail." 2-25: "I went from only knowing the 'what' to now knowing the 'why,' so I feel I understand the business much more deeply."
Code 08	Competitor and external environment analysis	Respondents extend BI tools to monitor external markets and analyze competitors.	5-23: "I noticed that prices for my restaurant on Platform A are higher than competitors on Platform B, or competitors offer meal sets I don't have. This can be put on the BI system, if it significantly affects my transaction volume, as a salesperson I would immediately follow up with that merchant."

Code 09	Decision support and analytical justification	Respondents use data as an objective basis and persuasion tool for decisions, enhancing decision legitimacy.	2-13: "When you make a decision, you are no longer relying on gut feeling, the data is right there, making it look more scientific and persuasive. Work efficiency improves, and people don't argue as much after looking at the same data."
Code 10	Strategy iteration and A/B test	Respondents use A/B experiments to systematically validate strategic hypotheses, treating data results as iteration basis.	11-13: "We conduct A/B experiments to select the best copy presentation and optimization for external display."
Code 11	Experience, intuition, and subjective judgment	Respondents describe ongoing dependence on experience and intuition during the data analysis process.	1-14: "Our decisions are not purely rational — we also integrate the experience and value judgments of business leaders." 6-30: "Data helps us make decisions efficiently, but it doesn't necessarily make our decisions better — decisions ultimately depend on people using their business experience."
Code 12	Data-based validation and the overturning of prior assumptions	Respondents' coping mechanisms when data conclusions conflict with existing experience.	1-7: "If data conflicts with experience, we usually run a more rigorous experiment or look at longer-cycle effects. Leadership also encourages bold assumptions and careful verification." 2-9: "Sometimes the data structure in the BA system differs from my leader's experience-based judgment, and the leader first questions whether the data calibration is wrong or the dimensions are off, requiring me to verify repeatedly."
Code 13	Data detached from the real business context	Respondents criticize data analysis or BI tool design that is detached from actual business needs.	7-23: "Many of our data product developers don't understand the business — they just take requirements and implement them, without knowing what the thing is for." 9-15: "Upper management trusts BI-produced dashboards more than they trust us (the business side), creating

			cognitive misalignment."
Code 14	Data-driven business logic	Respondents describe the process by which data thinking becomes internalized as the underlying logic of business operations.	2-1: "As a business operations person, my daily work involves user growth — campaigns, user segmentation — essentially driven by data throughout." 14-24: "People quickly look at the data; after doing so repeatedly, before making a decision they habitually check the numbers and perform preliminary data analysis. These data capabilities become embedded in them."
Code 15	The allocation of final decision-making authority	Respondents describe the functional boundary between data and business teams in the decision chain.	1-8: "We align on metrics together; the data team delivers dashboards or reports, and the business side makes decisions — decision authority rests with the business owner." 4-25: "It's like we're the rider on the pig, holding the carrot and extending it forward or backward, while the business team is the pig running forward — they don't need to think much, just follow the red line we provide."
Code 16	Voice Advancement and Role Progression	Respondents describe how mastery of data capabilities and BI tools has reshaped their authority within the organization.	1-20: "Before, when data was opaque, our role was more like a data-fetching tool. Now with BI, the business side can see 80% of daily data themselves, and our role has become explaining, judging, and advising — like data advisors." 9-22: "Before I was an executor; now I feel I've become a co-creator, a collaborator involved in projects, even taking on the identity of a business manager."
Code 17	Complementarity between business experts and data experts	Respondents describe the collaboration between business experts and data experts as a complementary division of skills.	1-24: "(Business side as students) The data expert provides vision, evidence, and risk warnings but does not make decisions for the business —

			like a teacher who provides resources but doesn't take the exam for you." 2-23: "Data experts build the tools accurately and provide reliable data support; business experts know the user groups and tell data experts what we actually need to look at."
Code 18	The barrier created by hands-on business experience	Respondents describe the hands-on experiential barriers possessed by business experts that data capabilities alone cannot replace.	7-29: "If you want to advance in data analysis as a career, you definitely need to understand the business — even better if you've worked in it for a while." 10-28: "Business experts also have certain barriers, including commercial negotiation, coordinating key interest decisions, and on-the-ground implementation."
Code 19	Cross-department communication and collaboration frictions	Respondents describe process-based division of labor between data and business teams in daily collaboration and friction arising in work cycles.	3-9: "I asked the data department to build it, but they have a different perspective — they feel the data is difficult to handle and friction arises over progress." 15-18: "When I communicate with them, I feel they don't understand the specific process of my business, leading to minor conflicts — but this is easy to resolve; once I provide the data, they stop fixating on the specifics."
Code 20	Differences in perspective between management and operation teams	Respondents describe structural gaps in data needs, metric focus, and technical understanding between management and frontline execution.	5-9: "As a frontline business person, I only care about the small slice of my business — if I'm a salesperson, I only care about the metrics tied to my bonus; everything else I'm not much interested in." 12-18: "Upper management's focus may be less comprehensive but more core — they find the key. As a module lead, you look at the data from its entire history and across all chains."

Code 21	Data tools and metric system construction	Respondents describe the composition of the internal data tool system and the practice of metric construction.	5-16: "The drill-down dimensions are based on analysts' past successful analysis experience — dimensions where anomalies are likely to appear or where there are actionable levers — and we include them directly as BI drill-down dimensions." 5-22: "Generally when building dashboards, the core metrics will have two or more dashboards for cross-validation."
Code 22	Base data governance and disorder	Respondents describe systemic disorder in the underlying data infrastructure of organizations.	4-3: "Our company's data foundation is very chaotic — we introduced Netease BI in 2023 to replace old systems, but it couldn't fully replace them, so we still use both platforms today." 4-17: "Every person has their own table — subsidiaries have different departments, claims has claims tables, product teams have product tables with fields they can't put on record but can't ignore, so there's a lot of gray area." 5-24: "If a company's infrastructure or underlying data has problems, the BI will be highly misleading and delay business actions."
Code 23	Resource redundancy and system overhead	Respondents describe resource mismatches and efficiency losses in the development and maintenance of data tools.	1-11: "Dashboard proliferation — everyone needs their own dashboard but actual usage frequency is low; the maintenance cost is very high." 5-28: "In reality, many analytics teams build piles of dashboards, but the download count or daily view count is nearly zero."
Code 24	Difficulties in metric alignment and consensus building	Respondents describe persistent friction caused by inconsistent metric definitions in cross-departmental collaboration.	10-16: "The biggest issue is metric calibration — at the data source level and for aggregate or complex calculated metrics, if there's no consensus, discussions about how to understand a metric can go on and on."

			11-19: "The most common issue is metric calibration — if unclear, it can go back and forth for a long time, because the same metric with different statistical scopes may yield different results."
Code 25	The cost of data explanation and user education	Respondents describe the extra explanatory burden created by differences in data literacy.	12-19: "When a number is unreliable due to statistical fluctuations or sample size issues, your superior often doesn't know the reason, so you spend a lot of time explaining why you can't just look at one number or metric." 13-16: "Everyone understands and interprets data differently, so there are misunderstandings, gaps, and sometimes conflicts."
Code 26	Formalism and data decoupling	Respondents reveal a systemic phenomenon where data is instrumentally used to serve predetermined conclusions, maintain surface-level healthy metrics, or project a 'data-driven' image.	4-29: "There's a well-known saying: 'There's no wrong data, only wrong calibration' — if you think you've crossed the red line, just change the calibration and it disappears." 13-22: "We found that a large portion of walk-in customers came through channel vouchers, but to hit their KPIs, agents brought in people with no real purchase intent."
Code 27	Risk avoidance and transfer of responsibility	Respondents describe survival strategies used under performance pressure to avoid risk and shift responsibility through data manipulation.	3-7: "If an issue persists unresolved, we may issue fines or escalate to cross-departmental communication." 4-30: "We don't manage data for efficiency — just to play with numbers and maintain apparent health in front of headquarters. It's really about wrapping the bomb layer by layer and passing it down the line, making sure it doesn't explode in front of this leader or that leader."
Code 28	Job responsibilities and boundary definition	Respondents describe their own role positioning and functional boundaries.	15-1: "I'm a big data development engineer — my main work is supporting the company's BI business development and maintenance, under the data middle platform

			team."
Code 29	The marginalization and replacement of basic work	Respondents describe how BI tools and automation processes are systematically replacing basic data operation roles.	11-33: "It may marginalize some more junior positions, but new roles will also emerge — emphasizing operations staff who understand data, rather than purely mechanical operators." 15-30: "Before, there were at least 3 people online; now it's just me and one intern, because traditional manual reports have been fully replaced and marginalized by updated tools like ITR and BI reports."
Code 30	The emergence of demand for hybrid roles	Respondents describe how BI tool proliferation and digital transformation are creating new hybrid role requirements.	14-28: "Analysts are no longer just doing analysis and visualization — they increasingly need to communicate with AI and train AI." 15-32: "We're preparing to hire a new role focused on data governance, compliance, and export management — creating new positions like these."
Code 31	Upgrading soft skills and communication ability	Respondents highlight communication and narrative capabilities overtaking pure technical skills following role reconfigurations.	11-31: "A high-level operator basically needs to be proficient in both business and data — only then is their work more convincing and precise." 13-33: "The focus has shifted — no longer staying on the surface of data, but deeply understanding the story behind the data. This raises the bar for analytical capability."
Code 32	Career anxiety in the face of AI substitution	Respondents express career threat anxiety triggered by AI technology entering data analysis work.	6-35: "If AI integration can highly efficiently optimize BI data and calculations, I believe a lot of manpower can be cut." 14-27: "All professional knowledge and experience needs to be fed into AI — that gives me some anxiety. After an analyst's experience is given to AI, will the analyst's value

			diminish?"

c. Informant Profile

ID	Job Title	Decision Role	Industry	Company Size	BI Experience
S01	Digital Product Manager	Platform Builder / Data Advisor	Internet	10,000+	Heavy / Architecture-Level User / Heavy (Developer & Analytics Expert)
S02	Internet Business Operations	Demand Initiator & Decision Advisor	Internet / E-commerce	1,000+	Heavy (Business Loop-Driven) / Hands-on
S03	UX Product Manager	Demand Initiator & Stakeholder Manager	Internet	1,000+	Moderate / Project-Based
S04	BI Data Analyst	Defensive Decision Supporter / Metric Controller	Traditional Finance / Non-Auto Insurance Claims	10,000+	Moderate / Reluctant Adopter
S05	Senior Business Analyst (BA)	Organizational Bridge & Third-Party Observer	Internet / Multi-Business Platform	10,000+	Deep / Architecture-Level User
S06	Business Analyst	Middle-Platform Decision Supporter	Internet / F&B Supply Chain	10,000+	Moderate-Heavy (Analysis-Oriented)
S07	Middle-Platform Operations Analyst	Goal Setter & Solution Provider	Internet Live-Stream E-commerce	10,000+	Heavy / High-Frequency User
S08	Head of Risk Control	Risk Gatekeeper / Decision Participant	FinTech	300+	Heavy / Self-Service User
S09	BI Product Manager	Operations Decision Maker / End-to-End Owner	Whole-Home Intelligence	10,000+	High-Frequency / Hybrid User
S10	Business Analyst	Analytics Middle-Platform & Joint Decision Maker	Internet Mobility Platform	10,000+	Heavy / Expert-Level
S11	Product & User Operations	Business Decision Maker & Dashboard Builder	Internet Mobility Platform	10,000+	Heavy (Hands-on)
S12	Strategy Lead	Business Decision Maker & Demand Definer	Internet Mobility Platform	10,000+	Heavy (Strategy Monitoring Perspective)
S13	Data Operations Manager	Decision Supporter / Data Interpreter	Real Estate	10,000+	Heavy (Senior Analytics Perspective)
S14	Data Analyst	Decision Supporter & Tool Builder	Internet	10,000+	Heavy / Expert-Level
S15	Big Data Development Engineer	Technical Builder / Decision Supporter	Retail	300+	Heavy / Developer & Monitor